

Company Websites: A New Measure of Disclosure

ROMAIN BOULLAND ^{*}, THOMAS BOURVEAU [†],
AND MATTHIAS BREUER [‡]

Received 31 March 2021; accepted 15 July 2024

Abstract

We propose a new measure of firms' disclosure based on company websites, which are widely available and contain a wealth of information. For a sample of U.S. public firms, we construct our disclosure measure using historical website data, validate it by correlating it with extant measures of disclosure and information asymmetry, and explore its determinants. We then apply our measure to the study of U.S. private firms' disclosure and French firms' compliance with a nonfinancial disclosure mandate. Our applications illustrate that our website-based measure provides a useful complement to extant

^{*}ESSEC Business School; [†]Saïd School of Business, University of Oxford; [‡]Goethe University Frankfurt.

Accepted by Rodrigo Verdi. We thank Matt Bloomfield, Baptiste Colas (discussant), Ed deHaan, Henry Friedman, Mirko Heinle, Philip Joos, Christian Leuz, Bill Mayew (discussant), Anya Nakhmurina, Shiva Rajgopal, Delphine Samuels, Bryce Schonberger (discussant), George Serafeim, John Wertz (discussant), Christina Zhu, seminar participants at Harvard Business School, London Business School, HEC Lausanne, and conference participants at the 2019 Stanford Summer Camp, the 2019 Dartmouth (Tuck) Accounting Conference, the 2019 Burton Conference at Columbia University, the 2021 Conference on Financial Economics and Accounting (CFEA), the 2022 Financial Accounting and Reporting Section (FARS) Mid-Year Meeting, and the 2022 European Accounting Association (EAA) Annual Conference for helpful discussions and comments. We thank Rongcheng Li for his outstanding work as a research assistant on this project. We thank Marlene Plumlee and Jing He for sharing their 8-K measures, and Shuping Chen for sharing her disclosure quality measure. Data and code to construct our measure are available on our GitHub repository (github.com/r-boulland/Company-Website-Disclosure). The authors have no conflict of interest to declare. An online appendix to this paper can be downloaded here: <https://www.chicagobooth.edu/research/chookaszian/journal-of-accounting-research/online-supplements-and-datasheets>

measures of disclosure, which are more narrowly focused on investors in public capital markets.

JEL codes: D82; G14; G32; G34; K22; M10; M30; M40

Keywords: websites; disclosure; private firms; private equity; ESG

1. *Introduction*

Firms have incentives to communicate information to their various stakeholders, including investors, suppliers, customers, and employees (e.g., Beyer et al. [2010], Dranove and Jin [2010]). In recent years, this communication has shifted toward and been facilitated by the Internet. The Internet allows firms to directly communicate with their stakeholders without costly intermediation, and to reach all their various stakeholders at once. As a result, practitioners view the Internet, and firms' websites in particular, as a powerful public disclosure channel for both public and private firms (e.g., IR-Society [2012], Farmiloe [2023]). To date, however, most studies focus on disclosure channels other than company websites, such as regulatory filings or conference calls. Those channels are routinely used by public firms to communicate with their investors, but they are less commonly used by firms that are not publicly listed and firms that seek to communicate with audiences other than investors. As a result, the literature has lost sight of disclosures initiated by private firms and targeted at noninvestor audiences. This is an increasingly glaring omission given the important role private firms play in the economy and today's pressing environmental, social, and governance (ESG) issues (e.g., Minnis [2022], Gözlügöl and Ringe [2022]).

In this paper, we develop a new measure of firms' public disclosure based on their company websites. This website-based measure promises to be particularly useful for research on private firms and the disclosure of nonfinancial information to various stakeholders.¹ In the first part of this paper, we describe the construction of our website-based measure, validate it as a measure of disclosure, and explore its determinants. To facilitate the comparison to prior literature, we focus on a sample of public firms in the United States (U.S.). For these firms, other disclosure measures are widely available. In the second part, we showcase the usefulness of our measure by applying it to two settings where extant measures are lacking: disclosure by private firms and disclosure of nonfinancial information. In the first application, we show that our measure can be constructed for millions of private firms in the U.S. In the second application, we show that our measure can be customized to measure and compare private and public firms' responses to nonfinancial disclosure mandates, specifically, a mandate in France requiring the disclosure of firms' Gender Equality Index scores on firms' websites. These applications help illustrate the sheer

¹ For private firms and for nonfinancial information, websites constitute an important channel—if not the main channel—of public disclosure (e.g., Leonelli et al. [2025]).

scale of the gap that our measure helps fill, the versatility of our measure, and the importance of expanding our focus beyond public firms.

We obtain data on firms' historical websites from the Wayback Machine, a digital archive maintained by the Internet Archive. The Wayback Machine periodically crawls and stores snapshots of nearly all existing websites. It provides a list of all Uniform Resource Locators (URLs) constituting a website, the size (in bytes) of each URL, and whether a specific URL contains only text or enriched content such as images, videos, or specific applications (e.g., PDF files). For the first part of our paper, we obtain this information for all U.S. public firms with nonmissing URLs in COMPUSTAT on a quarterly basis, from Q1 1997 to Q4 2020. For the second part, we also obtain website information for millions of U.S. private firms and select French firms with nonmissing URL information in either Bureau van Dijk's Orbis or COMPUSTAT Global.

We measure firms' website-based disclosure using the total size of firms' websites. This measure conceptually captures the quantity of information provided to stakeholders. In this regard, it is similar to existing disclosure measures capturing the length of firms' filings (e.g., Dyer, Lang, and Stice-Lawrence [2017], He and Plumlee [2020], Breuer, Hombach, and Müller [2023]) or the number of disclosed items (e.g., Lang and Lundholm [1993], Botosan [1997], Chen, Miao, and Shevlin [2015]). In addition to this broad website-size measure, we also construct two types of content-specific measures. Our first content-specific measure is based on the information contained in firms' URL strings. This measure allows us to examine broad content categories discussed on firms' websites: e.g., products, investors, human resources, and geography. Our second content-specific measure is based on textual analysis of firms' entire website content (e.g., a full-text search for ESG-related content).

We validate our website-based measure by correlating it with extant measures of firms' disclosure and investors' information asymmetry. We find that our website-based measure is positively associated with four popular measures of public firms' (voluntary) disclosure: the number of management earnings forecasts (Marquardt and Wiedman [1998]), the extent of disaggregation in firms' financial statements (Chen, Miao, and Shevlin [2015]), the word count of voluntary 8-K filings (He and Plumlee [2020]), and the gross 10-K file size (Loughran and McDonald [2014]).² In addition, we find that our website-based measure is negatively associated with investors' information asymmetry as measured by public firms' bid-ask spreads. Collectively, these results indicate that our website-based measure captures public information that is relevant to stakeholders such as investors.

² For our 8-K filings measure, we use the word count of voluntary 8-K filings. He and Plumlee [2020] suggest that this measure is superior to measures based on the count of filings.

We next explore the determinants of our website-based measure of firms' disclosure. Using the sample of U.S. public firms, we document that firms' websites contain mostly text, though images and other media have been increasing in importance in recent years. With respect to website contents, we find that our public sample firms discuss central aspects of their business on their websites. The two most common topics include firms' products, strategy, and processes (PSP) and investor relations (IR). We also find that firms' website size, components, and contents vary systematically with firm characteristics. In the cross-section, for example, we find that larger firms, firms operating in business-to-business industries, and firms with many analysts, high institutional ownership, and volatile businesses disclose more information on their websites. In terms of components and contents, we observe that larger firms employ more applications on their websites and talk more about non-PSP contents, including their human resource practices. While individual firm- and industry-level characteristics have some explanatory power, we observe that most of the variation in firms' websites, especially their size and contents, appears to be driven by time-invariant firm effects. This pattern is consistent with firms adopting and largely committing to a particular website disclosure strategy. Still, a notable share of variation in firms' website sizes and contents remains unexplained by time-invariant fixed effects and broad time trends. Accordingly, our website-based measures allow researching both across- and within-firm forces driving managers' incentives to disclosure to the public.

Finally, we present two applications of our measure to showcase its usefulness for research on private firms and disclosure of nonfinancial information. Private firms make up the vast majority of firms and account for a substantial share of economic activity in the U.S. and elsewhere. They are also of great importance in today's environmental and societal debates (e.g., Gözlügöl and Ringe [2022]). Still, the actions of private firms are often more opaque than those of public firms (e.g., Lino et al. [2022]). In the U.S., for example, most private firms do not have to disclose their financial information publicly (Minnis and Lisowsky [2020], Crenshaw [2021]). As a result, most accounting research in the U.S. focuses on the readily available disclosures of public firms. It remains unclear, though, to what extent insights gained from public firms carry over to private firms. With the following applications, we illustrate how our measure can be used to extend the literature beyond the *public* capital-market context (i.e., public firms and financial disclosures).

In our first application, we explore the determinants and capital-market outcomes associated with U.S. private firms' website disclosure. We construct our website-size measure for more than 2.8 million private firms for which other measures of disclosure are absent. For this large sample, we observe that the contents of private firms' website disclosures broadly align with those of public firms' disclosures (e.g., across industries), though private firms predictably talk less about IR than public firms. We also observe that private firms' size relates positively to their website disclosure, just as in the case of public firms. Most notably, we find that U.S. private firms'

website disclosure is positively associated with private equity transactions, even after we control for firm size. This finding provides the first large-scale evidence consistent with U.S. private firms' voluntary (website) disclosure increasing their equity's liquidity. Even in the absence of regulatory filings (e.g., on SEC's EDGAR), private firms appear to engage in public disclosure on their websites (e.g., to attract private equity investors and other stakeholders; Breuer, Hombach, and Müller [2023], Baik, Berfeld, and Verdi [2025]). This finding also suggests that the relation between public disclosure and firms' equity liquidity may extend beyond the public firm setting to millions of private firms.

In our second application, we explore how private and public firms respond to a recent nonfinancial disclosure mandate in France. Since 2019, public firms and larger private firms have been required to submit a Gender Equality Index to the Department of Labor and to publish the score on their websites. Using the Wayback Machine, we construct customized measures of private and public firms' compliance with this disclosure mandate. We find that private firms are significantly less likely to comply with the mandate. Most notably, we find that private firms' (non)compliance is strongly related to their underlying Gender Equality Index, which is available through the Department of Labor. We observe that private firms with higher gender inequality are less likely to report the index on their websites, and if they do report it, they show it on less prominent sub-pages. Interestingly, this strategic disclosure behavior is observed only for private firms. The limited noncompliance of public firms appears unrelated to their workforce's gender composition and practices. This analysis highlights that private and public firms can respond differently to disclosure mandates, which suggests the need for caution in generalizing findings from public firms to private firms. It also underscores the importance of extending our focus beyond public firms, especially as disclosure of nonfinancial information is increasingly used as a tool to address ESG issues.

Our paper contributes a new website-based measure of firms' disclosure to the literature. Prior measures capture several dimensions of firms' disclosure, including the quantity (Dyer, Lang, and Stice-Lawrence [2017]), disaggregation (Chen, Miao, and Shevlin [2015]), frequency (Marquardt and Wiedman [1998], He and Plumlee [2020]), and precision (King, Pownall, and Waymire [1990], Hutton, Miller, and Skinner [2003]) of disclosures such as SEC filings and management forecasts. While our website-based measure captures similar dimensions of firms' disclosure, it adds to the extant measures in important ways. It is more broadly available and targeted at a wide range of audiences, not just investors in public capital markets. These features make our measure particularly useful for research on private firms, noninvestor audiences, and nonfinancial disclosures. Future research can explore both voluntary disclosure on websites as well as responses to disclosure mandate that increasingly consider websites as a disclosure channel (see our French private setting example and the disclosure mandate pertaining to supply chain for firms doing business in California studied by She [2022]).

Our paper also contributes to the literature on information-dissemination channels. Prior literature investigates the role of various channels, including the news media (Bushee et al. [2010], Rogers, Skinner, and Zechman [2016]), social media (Blankespoor, Miller, and White [2014], Blankespoor [2018]), SEC's EDGAR (Christensen, Floyd, and Maffett [2017], Rogers, Skinner, and Zechman [2017]), and Wikipedia (Zhu [2019]). Firms' websites have received less attention, although firms and regulators increasingly turn to websites as disclosure outlets. Prior work on website disclosure provides in-depth descriptions of the amount and content of information provided on websites for small samples of public firms (e.g., Koreto [1997], Ettredge, Richardson, and Scholz [2001], Matherly and Burton [2005], Cormier, Ledoux, and Magnan [2009], Pollach [2011], Saxton, Neely, and Guo [2014], Pownall, Vulcheva, and Wang [2015], Arora and Lodhia [2017]). In addition, recent work by Levy and Taylor [2022] documents that public firms' websites contain financial information that is useful for investors.³ Our paper contributes to this stream of the literature by providing a large-sample exploration of firms' website disclosure, which includes disclosures of private firms and nonfinancial information (e.g., ESG content).

Our paper is related to other streams of literature as well. It relates to the growing literature on private firms' disclosure. Prior work examines the consequences of private firms' mandatory disclosure, typically in the European setting (Bernard [2016], Breuer, Hombach, and Müller [2017], Breuer [2021], Aghamolla and Thakor [2022], Ortiz et al. [2023], Baik, Berfeld, and Verdi [2025]). Our evidence adds novel, large-scale evidence on private firms' voluntary disclosure and its relation to private equity markets in the U.S. It complements recent survey evidence by Kang [2025] on private firms' communication with equity investors for a small sample of U.S. private firms, and recent evidence by Kim and Olbert [2022] on the consequences of private firms' disclosure for private and public capital markets. Our paper also relates to the burgeoning literature using textual analysis of text corpora to measure firms' activities and exposure. Prior work, for example, uses regulatory filings and conference call transcripts to measure firms' political risk, regulatory intensity, or climate exposure (e.g., Hassan et al. [2019], Andreicovici et al. [2020], Sautner et al. [2023]). Our paper highlights the promise and advantages of using text provided on company websites. In line with our paper, Hoberg et al. [2024] take advantage of the broad coverage of company websites to create more accurate and comprehensive text-based measures of industry competition and classification. Lastly, our paper relates to the growing literature making use of the Wayback Machine in various contexts. Several papers use the Wayback Machine

³Levy and Taylor [2022] note that since 2008, publicly listed firms in the U.S. have been permitted to substitute their mandatory financial information disclosure on EDGAR with corresponding disclosures on their company websites. Consistent with our paper, they document that company websites contain useful information for investors.

to obtain historical information from the websites of firms, investors, or regulators.⁴ Our paper illustrates how researchers can use the Wayback Machine to construct a broad historical panel of firms' websites, which allows them to study various aspects of firms' activities, including their disclosure.

2. Construction

In this section, we describe the construction of our website-based measure of disclosure. We discuss the source of website data, the sample of firms for which we obtain the website data, the disclosure measures that we construct from the website data, and the variation and coverage of the resulting measures.

2.1 WAYBACK MACHINE

We obtain full historical records of company websites from the Wayback Machine, a digital archive of websites maintained by the Internet Archive.⁵ Since 1996, the Internet Archive, a nonprofit organization financed by various foundations and individual donations, has aimed to preserve digital contents of all sorts in a context of fast technological change. To this end, it has created a digital record of websites and other digital contents (e.g., software, movies, music) similar to a print library. It currently maintains more than 45 petabytes of data on various servers across the U.S. As part of its mission, the Internet Archive grants open access to the stored data.

As one of the main services of the Internet Archive, the Wayback Machine allows users to track the evolution of websites over time. Its algorithm periodically crawls the Internet to keep a record of all existing and newly created websites. While the specific crawling algorithm is confidential, the Wayback Machine provides some information on the reasons for website crawls. Most websites are crawled through a general sweep of the Internet. Some websites are crawled more frequently because they are top domains or otherwise of particular interest. The Wayback Machine also crawls specific websites in response to news events. The resulting records are snapshots of almost all existing websites at different points in time, accessible via the Wayback Machine's user-friendly interface. This interface prompts users to enter a web URL and allows them to navigate into the historical website as recorded at the chosen point in time (see the time banner on the screenshots in figure 1).

⁴ Prior studies have relied on the Wayback Machine to retrieve various types of information from historical snapshots of websites, including lists and locations of regulators' offices (e.g., Lim, Hagendorff, and Armitage [2016], Mas [2017], Ege, Glenn, and Robinson [2020]), lists of asset-pricing factors from financial intermediaries (Akey, Robertson, and Simutin [2023]), timing of firm-specific events such as amendments to governance guidelines (Bebchuk and Tallarita [2022]) or adoption of technologies (Buchak et al. [2020]), and the timing of market penetration for new products/services such as Craigslis (Gao, Lee, and Murphy [2020]).

⁵ <https://archive.org/web/>

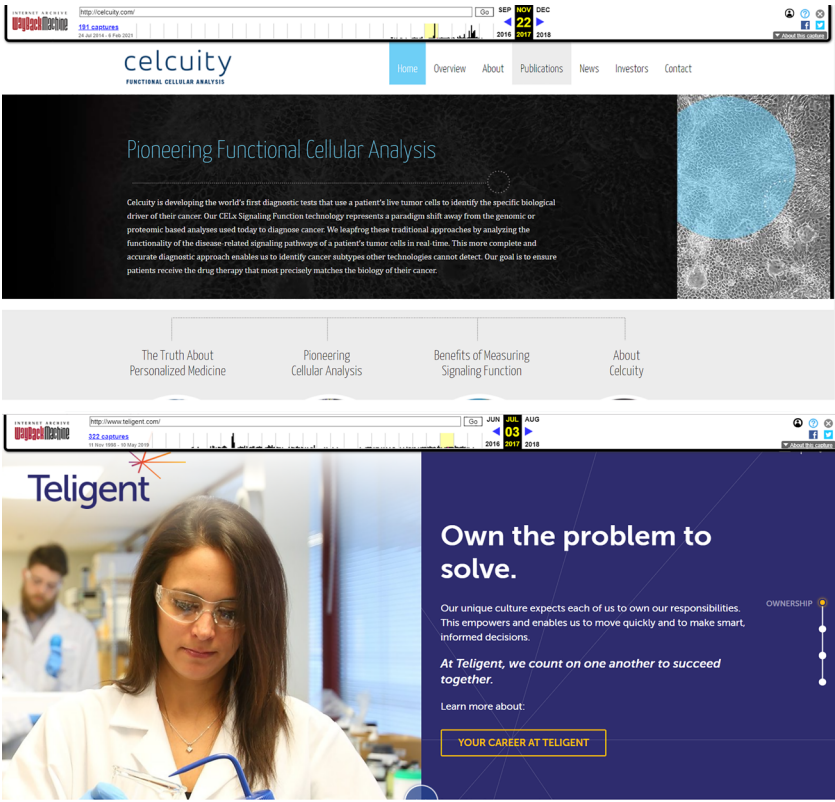


FIG. 1.—Examples. The figure shows screenshots from the fourth quarter of 2017 of the main landing company webpages of two pharmaceutical firms included in our sample, *Celcuity Inc.* and *Teligent Inc.*

In addition to the user-friendly interface, the Wayback Machine provides access to the underlying archive data through an Application Programming Interface (API) that allows users to collect the historical structure of the website. For each archived website, the API provides access to the list of all URLs constituting the website (variable “urlkey”), a “timestamp” variable indicating the day at which a URL exists, the size (in octets or bytes) of each URL, and a “status code” variable indicating whether a URL points to a webpage or the query results in an error (e.g., 404 for the “404 not found” error). In addition, the “mimetype” variable indicates whether a specific URL contains text only (e.g., “text/html”), or enriched content such as images, videos, or specific applications (e.g., PDF files).

2.2 SAMPLE

We obtain snapshots of company websites for public and private firms. In the first part of this paper, we focus on a sample of public firms to illustrate the construction of our website-based measure, describe its

variation, and validate it as a measure of firms' disclosure. We describe this sample below. In the second part, we expand our focus to private and international (i.e., French) firms. We describe those samples in the applications section (section 5). An overview of the sample selection is provided in appendix A.

For the U.S. public firm sample, we start with all firms in COMPUSTAT U.S. with nonmissing company website URLs over the period 1997-2020 (COMPUSTAT variable "weburl"). This step results in 13,339 firms. We collect historical data on each website (an archive) from the Wayback Machine API by using queries of the following form: <http://web.archive.org/cdx/search/cdx?url=www.cecoenviro.com&matchtype=domain&collapse=timestamp:10&matchType=prefix&output=json>

In the example query above, we ask the Wayback Machine API for all records of the website of CECO Environment. We obtain historical records from [cecoenviro.com](http://www.cecoenviro.com) (the main domain) as well as all its subdomains (by setting "matchtype = domain"). This query returns a list of records (rows), each one representing a URL. The results are provided in JSON format, a light format that is well suited for the storage of large amounts of data. Following these steps, we are able to download 12,800 archives of company websites (96% of the initial dataset), amounting to about 155 Gigabytes of data.

We read each JSON file and remove URLs whose "status code" corresponds to an error, and we extract the date of each record from the "timestamp" variable. A given URL may appear several times in the database, corresponding to each time the website was crawled by the Wayback Machine algorithm. We first collect all the records available for each firm. To construct disclosure measures on a quarterly basis, we compute the average values for that URL in a given quarter whenever there are several records of a URL available in a given quarter. If there are no records available in a given quarter, we code this observation as missing. This coding prevents us from conflating website characteristics that are under the control of firms (e.g., their website size) with the frequency of snapshots taken by the Wayback Machine, which is largely outside of the control of firms.

We keep firms for which we can compute capital-market variables (i.e., quarterly bid-ask spreads, market values, share turnover, and return variability) by merging our dataset with the CRSP-COMPUSTAT merged database. This step results in a sample of 5,234 unique U.S. public firms corresponding to 198,316 firm-quarter observations over the 1997-2020 period. We add four commonly used measures of disclosure to this sample: the number of management earnings forecasts (Management Forecast), the number of disaggregated items in firms' financial statements (Disclosure Quality, period), the word count of voluntary 8-K filings (Voluntary 8-K Filings, period), and the number of characters in the original 10-K filings (Gross 10-K File Size).

2.3 MEASURES

2.3.1. Website Size. We construct our main measure of website-based disclosure by taking the logarithm of firms' website size (*Website Size*: the total size of the website in bytes). This simple measure allows for a standardized comparison across firms and over time. Conceptually, it captures the quantity of information or level of detail provided via firms' websites, in line with previous disclosure measures (e.g., Loughran and McDonald [2014], Chen, Miao, and Shevlin [2015], He and Plumlee [2020]). While firms can elect to provide mandatory disclosures such as SEC filings on their websites (e.g., Levy and Taylor [2022]), the vast amount of variation in firms' website size represents firms' voluntary disclosure choices.

To illustrate the variation captured by our website-based disclosure measure, we present two examples of company websites in figure 1. We show screenshots of the landing pages of two firms' websites as captured in Q4 of 2017. To facilitate comparison, the two example firms, *Celcuity Inc.* and *Teligent Inc.*, operate in the same industry (i.e., pharmaceuticals), are listed on NASDAQ, and belong to the third decile of market capitalization. According to our standardized measure, *Celcuity Inc.* had a relatively small website in 2017 with a size of 449 kilobytes. The website is fairly generic and contains only a brief description of the firm's products and standard financial information. By contrast, our measure shows that the *Teligent Inc.* website was much larger in 2017, with a size of 11,294 kilobytes. It contained detailed information on various topics, including the firm's investments, manufacturing capabilities, and existing products. The "investing" section further contained a document detailing its plan for expansion and the job positions associated with this expansion. Browsing the two examples of historical websites illustrates that the relative size of the websites tends to reflect the amount of information the sites provided.

2.3.2. Website Content. We complement our standardized website-size measure with measures examining the content of websites. The content-based measures aid our understanding of the information disclosed on firms' websites and illustrate the potential for customization of our website-based measure for specific research questions (e.g., questions about product information or ESG issues). To construct content-based measures, we take three distinct approaches.

First, we construct disaggregated measures capturing the components (i.e., text, media, and applications) of each website and the specific applications used on the website (e.g., PDF, Word, Excel, Java) using the "mime-type" information. This disaggregation of firms' websites into components and application (or file) formats allows us to examine which parts of firms' websites provide outsiders, like investors, with useful information.

Second, we construct broad measures of website contents by parsing the words embedded in the URLs of firms' websites. Each URL in our database is a string that is potentially informative about the content of the webpage to which it points. For instance, a URL that contains the keyword "investor"

is likely to point to the investor-relation section of the website, while a URL containing the word “careers” likely points to job opportunities for prospective job applicants. We decompose the URLs into relevant content- and/or audience-related categories as follows. We purge all URL strings of the website host name. We retain all words in the URL appearing in a standard English dictionary.⁶ After purging stop words (e.g., pronouns), we obtain a list of 92,968 unique words. For each word, we compute the frequency with which firms use it in their URLs. This distribution of word frequencies is highly skewed because many words (23,020) are used only once. The words with low frequencies are typically highly specific words that are idiosyncratic to a firm’s business (e.g., cyclotron, photopolymer, or zebrawood). By contrast, the most frequent words tend to be programming-related terms that convey very little content-specific information (e.g., txt, robots, or contact). We remove words in the top 10% (used by more than 4,972 firms) and bottom 10% (used by less than 562 firms) of the distribution to focus our manual classification of URL content into relevant categories. This leaves us with 1,701 unique words to classify.⁷ We classify the words as belonging to broad content categories (e.g., “Investor Relations”) identified in two early exploratory studies on disclosures on company websites (Aerts et al. [2006], Cormier, Ledoux, and Magnan [2009]).^{8,9}

Third, we construct specific measures of website contents by analyzing the text on firms’ websites. For these measures, we need to query the Wayback Machine API to obtain all records of websites corresponding to the firms’ URLs, not just the meta data. Using the “timestamp” variable, we keep the last-seen record of each firm’s URL for each calendar quarter. This gives us a snapshot of company websites as they stand at the end of each calendar quarter. To analyze the content of these websites, we then proceed as follows: We first parse the host URL (i.e., the website main page) as well as its sub-URLs and store their content in a vector. We next clean the resulting parsed characters as follows: We remove irrelevant textual contents (e.g., HTML tags, Java scripts), numbers, and punctuation marks. Finally, we stem each word using the Porter algorithm.¹⁰ The

⁶ We use the following dictionary: <https://github.com/dwyl/english-words>.

⁷ We do not rely on statistical methods (e.g., Latent Semantic Analysis) to group words into categories as we are not interested in words that co-occur; rather, we seek to understand what broader content or audience a given word may belong to or speak to.

⁸ Aerts et al. [2006] categorize the disclosed content of websites of 56 European and North American firms. Cormier, Ledoux, and Magnan [2009] adopt the same scheme to categorize the website content of 189 listed Canadian firms.

⁹ We are able to manually classify 467 words out of the full list of unique words into the broad content (or audience) categories. The remaining 1,199 words cannot be used to identify the topic of the URL either because they are programming-specific (e.g., background, banner, color), or because they on their own do not convey sufficient information (article, chart, deliver, etc.).

¹⁰ The Porter stemming algorithm removes the common morphological and inflectional endings from words in English. For example, the Porter algorithm reduces *argue*, *argued*, *argues*, and *arguing* to the stem *argu*.

TABLE 1
Descriptive Statistics (U.S. Public Firms)

	Obs.	Mean	SD	P25	P50	P75
Website-based disclosure variables:						
<i>Website Size (Megabytes)</i>	198,316	12.773	36.306	.141	1.050	7.147
<i>Website Size</i>	198,316	13.649	2.905	11.859	13.864	15.782
Existing measures of disclosure:						
<i>Management Forecast</i>	121,486	1.342	0.545	0.693	1.386	1.609
<i>Voluntary 8-K Filings</i>	184,425	2.036	0.556	2.118	2.197	2.258
<i>Disclosure Quality</i>	121,372	0.481	0.071	0.432	0.468	0.536
<i>Gross 10-K File Size</i>	357,071	13.331	1.861	11.769	13.157	14.707
Capital market variables:						
<i>Bid-Ask Spread</i>	198,316	-5.968	1.630	-7.290	-6.140	-4.614
<i>Market Value</i>	198,316	13.089	1.925	11.673	13.0765	14.434
<i>Share Turnover</i>	198,316	-5.361	1.041	-6.013	-5.236	-4.623
<i>Return Variability</i>	198,316	-3.646	0.524	-4.034	-3.668	-3.285
Determinant variables:						
<i>Revenue</i>	37,672	4.834	2.087	3.589	4.965	6.244
<i>Employment</i>	37,672	0.587	1.898	-0.794	0.642	1.974
<i>Analyst Following</i>	37,672	1.864	0.941	1.386	1.946	2.565
<i>Institutional Ownership</i>	37,672	0.648	0.284	0.453	0.719	0.881
<i>ROA</i>	37,672	0.009	0.099	-0.001	0.012	0.033
<i>Leverage</i>	37,672	0.223	0.242	0.033	0.172	0.336
<i>Return variability (raw)</i>	37,672	0.03	0.019	0.017	0.024	0.036
<i>Segments</i>	37,672	0.624	0.715	0.000	0.000	1.386
<i>Geographical Complexity</i>	37,672	2.302	0.618	1.792	2.303	2.708
<i>Restructuring</i>	37,672	0.324	0.468	0.000	0.000	1.000
<i>Business-to-Business</i>	37,672	0.659	0.474	0.000	1.000	1.000
Website components variables:						
<i>% (HTML)</i>	36,527	0.452	0.382	0.087	0.334	0.897
<i>% (Media)</i>	36,527	0.208	0.288	0.000	0.044	0.350
<i>% (Applications)</i>	36,527	0.283	0.347	0.000	0.074	0.578
Website content variables:						
<i>% (PSP)</i>	34,918	0.480	0.331	0.175	0.481	0.776
<i>% (IR)</i>	34,918	0.377	0.316	0.089	0.318	0.593
<i>% (Other)</i>	34,918	0.144	0.227	0.002	0.041	0.174

This table presents summary statistics for the variables used in our tests involving our sample of U.S. public firms. Our website-based disclosure variable (*Website Size*) is measured quarterly over the period 1997–2020. *Management Forecast* is measured quarterly over the period 2000–2018. *Voluntary 8-K Filings* is measured quarterly over the period 2005–2016. *Disclosure Quality* is measured annually over the period 1980–2017. *Gross 10-K File Size* is measured quarterly over the period 1996–2016. *Capital market variables* are measured quarterly over the period 1997–2020. *Determinant variables*, *Website components variables*, and *Website content variables* are measured annually over the period 2000–2020. All continuous variables are truncated at the 1st and 99th percentile. All variables are defined in appendix B.

resulting dataset comes in the form of a list of words constituting each website in a given quarter. Equipped with those data, we use a bag-of-words approach to examine the frequency of specific words/content of interest.

2.4 VARIATION AND COVERAGE

Table 1 provides descriptive statistics for the public-firm sample. It shows that the average website size of U.S. public firms in our sample is about 12.8 megabytes. The distribution of firms’ website size is highly right skewed, as

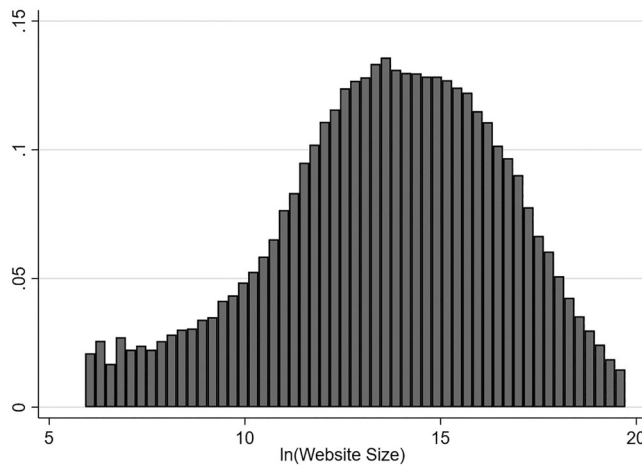


FIG. 2.—Distribution of *Website size*. The figure displays the distribution of our website-based measure of disclosure. *Website Size* is the natural logarithm of the (average) size of the company website in a given quarter. There are 229,562 observations from 5,215 unique firms over the period 1997–2020. We truncate *Website Size* at the 1st and 99th percentile.

is typical for size variables. To adjust for the skewness and reduce the impact of outliers, we use the natural logarithm of website size as our standardized measure of disclosure. Figure 2 plots this (logged) distribution. It shows that our measure exhibits substantial variation and that its distribution broadly follows a single-peaked bell shape.

Table 2 compares the coverage of our website-based measure for our sample of U.S. public firms with the corresponding coverage of extant disclosure measures (Management Forecast, Voluntary 8-K Filings, Disclosure Quality, and Gross 10-K File Size) over time. The comparison starts in 2005, the first year in which all five measures are available, and ends in 2018 when all other measures become unavailable. Relative to all firm-years in the CRSP–COMPUSTAT universe, our measure’s coverage is 54% in 2005 and reaches a high of 93% in 2018 at the end of the sample period. This gradual increase reflects the expanding coverage of the Wayback Machine over time. Moreover, it reflects that some firms’ website domains (main URLs) may have changed over time. As COMPUSTAT only reports the latest website domain, we construct our measure using the domain as reported in 2019, which inevitably contributes to missing observations and limited coverage in the earlier sample period.¹¹ Notably, this issue can be circumvented going forward by using annual snapshots of firms’ URLs provided on COMPUSTAT to track firms’ future website changes.

¹¹ We suspect that mergers and acquisitions also contribute to the low coverage at the beginning of our sample.

TABLE 2
Coverage of Disclosure Measures

Year	Full CRSP COMPUSTAT	Wayback Measure	%	Mgmt. Forecast	%	Vol. 8-K Filings	%	Disclosure Quality	%	Gross 10-K Size	%
2005	4,924	2,677	54.4%	2,383	48.4%	4,671	94.9%	3,169	64.4%	4,703	95.5%
2006	4,839	2,717	56.1%	2,485	51.4%	4,593	94.9%	3,089	63.8%	4,614	95.4%
2007	4,776	2,849	59.7%	2,399	50.2%	4,584	96.0%	3,014	63.1%	4,542	95.1%
2008	4,535	2,861	63.1%	2,216	48.8%	4,352	96.0%	2,916	64.3%	4,291	94.6%
2009	4,248	2,726	64.2%	2,220	52.3%	4,118	96.9%	2,243	52.8%	4,071	95.8%
2010	4,076	2,769	67.9%	2,190	53.7%	3,964	97.3%	2,041	50.1%	3,906	95.8%
2011	3,890	2,935	75.5%	2,236	57.5%	3,806	97.8%	1,941	49.9%	3,757	96.6%
2012	3,777	3,003	97.6%	2,193	58.1%	3,701	98.0%	1,857	49.2%	3,648	96.6%
2013	3,741	3,070	82.1%	2,073	55.4%	3,680	98.4%	1,809	48.4%	3,630	97.0%
2014	3,840	3,232	84.2%	2,017	52.5%	3,782	98.5%	1,840	47.9%	3,719	96.8%
2015	3,848	3,350	87.1%	1,944	50.5%	3,810	99.0%	1,848	48.0%	3,699	96.1%
2016	3,734	3,370	90.3%	1,782	47.7%	3,717	99.5%	1,800	48.2%	3,547	95.0%
2017	3,684	3,391	92.0%	1,802	48.9%	N/A	N/A	1,608	43.6%	N/A	N/A
2018	3,649	3,410	93.5%	1,722	47.2%	N/A	N/A	N/A	N/A	N/A	N/A

This table compares the coverage of our website-based disclosure measure with that of extant disclosure measures. We provide coverage of all disclosure measures relative to the full CRSP-COMPUSTAT universe of unique public firms. We provide statistics from 2005, which is the year in which all four extant disclosure measures start to become jointly available, up to 2018, when some of the extant disclosure measures become unavailable. This unavailability is caused by the absence of measure updates. All variables are defined in appendix B.

The coverage of our measure, especially in the latter years, is remarkably high. It compares favorably to the coverage of the disclosure measures based on management forecasts and disclosure quality. The coverage of the 8-K-based disclosure measure and the gross 10-K filing size is higher than the coverage of our website-based measures, but this higher coverage is driven in part by the fact that missing 8-K disclosures are coded as zeros.¹² It is also due to the fact that both SEC measures exploit variation in filings that are mandated for all the CRSP–COMPUSTAT firms in our sample.

Collectively, the descriptive statistics indicate that our website-based measure exhibits notable variation and is widely available. It exhibits substantial coverage advantages over several of the extant measures of disclosure, even among the sample of U.S. public firms. For private firms and nonfinancial disclosures, this advantage is even more pronounced, as we show in our later applications (section 5).

3. Validation

In this section, we validate our website-size measure as a measure of disclosure. For the sample of U.S. public firms, for which disclosure and information-asymmetry measures are widely available, we examine the association of our website-size measure with extant proxies for firms' voluntary disclosure and investors' information asymmetry. As a proper measure of disclosure, our website-size measure should be positively related to extant proxies capturing firms' disclosure decisions and negatively related to proxies for investors' information asymmetry (e.g., Diamond [1985], Diamond and Verrecchia [1991]).

3.1 VOLUNTARY DISCLOSURE

Table 3, panel A, presents univariate correlations between our website-size measure and the four extant disclosure measures, including management forecasts, voluntary 8-K filings, disclosure quality, and gross 10-K file size (e.g., King, Pownall, and Waymire [1990], Loughran and McDonald [2014], Chen, Miao, and Shevlin [2015], He and Plumlee [2020]). Our website-size measure is positively correlated with all four extant disclosure measures, providing *prima facie* evidence of the validity of our empirical proxies as measures of firms' public disclosure. To strengthen this evidence, we next turn to multivariate regressions, which allow us to account for potentially confounding differences across time periods, industries, and firms.

Table 3, panel B, presents results from regressions of our website-size measure on the four extant disclosure measures. In our baseline specification with year fixed effects, we find that all four disclosure measures are

¹²In 2019, we obtained disclosure measures based on regulatory filings developed in prior literature directly from the authors of these studies (e.g., Loughran and McDonald [2014], He and Plumlee [2020]). As a result, the coverage of these measures is limited to fiscal years before 2017 / 2018.

TABLE 3
Website Size and Extant Disclosure Measures

Panel A: Correlation matrix											
	Website Size	Management Forecast	Disclosure Quality	Voluntary 8-K Filings	Gross 10-K File Size						
Website Size	1										
Management Forecast	0.119 ^{***}	1									
Disclosure Quality	0.298 ^{***}	0.292 ^{***}	1								
Voluntary 8-Ks	0.115 ^{***}	0.043 ^{***}	0.044 ^{***}	1							
Gross 10-K File Size	0.267 ^{***}	0.095 ^{***}	0.534 ^{***}	0.215 ^{***}	1						

Panel B: Association between Website Size and extant measures of voluntary disclosure											
	Website Size										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Management Forecast	0.051 ^{***} (0.006)	0.040 ^{***} (0.005)	0.007 ^{**} (0.003)								
Disclosure Quality				0.841 ^{***} (0.083)	0.437 ^{***} (0.088)	-0.018 (0.073)					
Voluntary 8-K Filings							0.058 ^{***} (0.004)	0.059 ^{***} (0.004)	0.005 ^{**} (0.002)		
Gross 10-K File Size										0.030 ^{***} (0.002)	0.041 ^{***} (0.002)
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	No	Yes	No	No	Yes	No	No	Yes	No	Yes	No
Firm FE	No	No	Yes	No	No	Yes	No	No	Yes	No	Yes
Observations	76,979	76,979	76,979	29,090	29,090	29,090	111,023	111,023	111,023	156,167	156,167
Adjusted R ²	0.083	0.116	0.553	0.173	0.213	0.563	0.074	0.135	0.574	0.195	0.571

This table presents validation tests of our website-based measure (*Website Size*) as a measure of disclosure. In panel A, we provide pairwise correlations between *Website Size* and four extant measures of disclosure. In panel B, we present estimates of OLS regressions of *Website Size* on four extant measures of voluntary disclosure. The maximum sample period is 1997-2018. The effective sample period differs across each disclosure measure depending on the availability of the extant disclosure measure. The number of observations corresponds to the number of observations with overlap between our website-based measure and the respective extant measures and nonmissing observations for the four capital-market variables described in table 1. For readability, the coefficients on all four extant measures of disclosures are multiplied by 10. For brevity, the constant of the regressions is omitted from the tables. Standard errors, reported in parentheses below the coefficient estimates, are clustered at the firm level. ^{***}, ^{**}, and ^{*} indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in appendix B.

significantly positively associated with our website-size measure (columns 1, 4, 7, 10). When we include industry fixed effects, these associations attenuate for the relation with the management-forecasts measure and the disclosure-quality measure (columns 2 and 5), whereas they remain stable for the relation with voluntary 8-K filings and gross 10-K file size (columns 8 and 11). When we account for fine firm instead of broad industry fixed effects, the association between our website-size measure and the extant disclosure measures remains positive and significant for all measures but the disclosure-quality measure (column 6). The insignificant relation between our website-size measure and the disclosure-quality measure when we include firm and year fixed effects likely reflects the limited time-series variation in the disclosure-quality measure. By contrast, the significant relationship between our website-size measure and management forecasts, voluntary 8-K filings, and gross 10-K file size, even when we focus only on differential changes of firms' disclosures over time, plausibly indicates that our website-size measure captures meaningful changes in firms' disclosure decisions. In untabulated tests, we observe very similar results for the individual components of firms' website size (i.e., text, media, applications). These results suggest that firms use various components to disclose information (e.g., text, images, and PDF files). They also support our use of total website size as a summary measure of firms' website disclosure.

The positive association between firms' website size and extant disclosure measures suggests that our website-size measure captures similar variation as extant disclosure measures, validating our measure as a useful proxy for firms' disclosure. Indeed, our website-based measure appears to capture additional, independent variation, as the extant disclosure measures (and fixed effects) have limited power to explain our measure's variation (e.g., R^2 s range from 7% in column 7 to 57% in column 9). This variation may be related to disclosures geared toward stakeholders other than capital-market investors, the main audience of the four existing disclosure measures. This feature enables researchers to study distinct audiences (e.g., customers or regulators) and topics (e.g., ESG instead of financial performance), adding to the incremental usefulness of our measure over and above extant measures.¹³

3.2 INFORMATION ASYMMETRY

Table 4 presents results from regressions of bid-ask spreads on our website-size measure. Bid-ask spreads are a popular proxy for investors' information asymmetry (e.g., Lesmond, Ogden, and Trzcinka [1999],

¹³In the online appendix, we validate that our website-based measure can also be used to study content-specific disclosure (e.g., nonfinancial disclosure) in event time. Following Heflin and Wallace [2017], who show that oil and gas firms increased their environmental disclosures in 10-Ks after the Deepwater Horizon oil spill in 2010, we document that oil and gas firms also significantly increased their environmental disclosures on their websites after the spill (figure OA1 in the online appendix).

TABLE 4
Website Size and Bid-Ask Spreads

Panel A: Unconditional association between Website Size and Bid-Ask spreads					
	Bid-Ask Spread				
	(1)	(2)	(3)	(4)	(5)
Website Size	-1.944*** (0.045)	-0.612*** (0.021)	-0.092*** (0.017)	-0.056*** (0.017)	-0.064*** (0.014)
Market Value		-0.398*** (0.007)	-0.452*** (0.006)	-0.455*** (0.006)	-0.441*** (0.008)
Share Turnover		-0.521*** (0.009)	-0.434*** (0.007)	-0.425*** (0.007)	-0.250*** (0.006)
Return Variability		0.737*** (0.015)	0.506*** (0.014)	0.507*** (0.014)	0.230*** (0.011)
Year-Quarter FE	No	No	Yes	Yes	Yes
Industry FE	No	No	No	Yes	No
Firm FE	No	No	No	No	Yes
Observations	198,316	198,316	198,316	198,316	198,316
Adjusted R ²	0.120	0.680	0.806	0.808	0.872

Panel B: Association between Website Size and Bid-Ask spreads conditional on established disclosure measures				
	Management Forecast (1)	Discl. Quality (2)	Voluntary 8-Ks (3)	10-K File Size (4)
ln(Website Size)	-0.066*** (0.015)	-0.064*** (0.019)	-0.038** (0.013)	-0.073*** (0.016)
Established measure	-0.113*** (0.008)	0.177*** (0.034)	-0.017*** (0.006)	-0.016*** (0.004)
Observations	179,733	43,990	111,023	156,167
Adjusted R-square	0.873	0.866	0.913	0.878
Controls	Yes	Yes	Yes	Yes
Year-Quarter	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

This table presents correlations between our website-based measure of disclosure and bid-ask spreads. It reports estimates of OLS regressions of the natural logarithm of bid-ask spreads on *Website Size* for a sample of U.S. public firms. Panel A presents the unconditional association between Website Size and Bid-Ask spread. Panel B presents the conditional association after controlling for each established disclosure measure separately. In column (1) (Management Forecast) and (2) (Disclosure Quality), we set the disclosure measure to zero when it is missing. All control variables are lagged by four quarters. For readability, the coefficient on *Website Size* is multiplied by 10. Standard errors, reported in parentheses below the coefficient estimates, are clustered at the firm level. For brevity, the constant of the regressions is omitted from the tables. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in appendix B.

Leuz and Verrecchia [2000], Brüggemann et al. [2018]).¹⁴ In panel A, the unconditional association (no controls or fixed effects) of our website measure with firms' bid-ask spreads is significantly negative (column 1). This association attenuates but remains negative and statistically significant after we control for liquidity determinants (Christensen, Hail, and

¹⁴For brevity, we tabulate only the results using bid-ask spreads. Our untabulated results are largely robust to using a liquidity factor, as in Christensen, Hail, and Leuz [2013], that aggregates three liquidity proxies: zero returns, the Amihud illiquidity measure, and bid-ask spreads.

Leuz [2013]), including firm size (i.e., market value) (column 2). This pattern suggests that our measure, while positively correlated with firm size, does not merely reflect firms' size. The association further attenuates but remains negative and statistically significant after we control for time effects (column 3), industry effects (column 4), and firm effects (column 5).¹⁵ In untabulated tests, we find similar results when using the individual components (i.e., text, media, or applications) and contents (e.g., PSP or IR) of firms' websites. In panel B, we show that the negative relation between firms' website size and bid-ask spreads is robust to controlling for each of the extant disclosure measures individually. This finding suggests that, while our measure captures similar aspects as extant measures, it is not subsumed by any of the extant measures.

The negative association between firms' website size and bid-ask spreads suggests that our measure is negatively related to investors' information asymmetry, consistent with theory (Verrecchia [2001]) and prior disclosure measures (Coller and Yohn [1997], Chen, Miao, and Shevlin [2015], He and Plumlee [2020]). The negative relation appears to emerge because our website-based measure captures similar dimensions as extant disclosure measures but also (possibly) incremental information.

4. Determinants

In this section, we explore the determinants of our website-based disclosure measure. We first describe the components and contents that make up firms' websites and then examine factors associated with firms' website size, components, and contents.

4.1 COMPONENTS AND CONTENTS

Figure 3 plots the components of public firms' websites and their evolution over time. The upper panel documents that firms' websites consist primarily of text, followed by applications and media (e.g., images and videos). Over time, the proportion of media components increases. Among the applications, the lower panel documents that PDF files are the dominant application format used on public firms' websites, followed by Java and Flash formats. The relative frequency of the distinct formats appears to fluctuate over time, with some evidence of a substitutive pattern between PDF and Java formats.

Figure 4 shows the content categories discussed on firms' websites. The content categories are assigned based on the URL string content. The "Product, Strategy and Processes" (PSP) content category comprises URLs

¹⁵ The negative association is also evident in the subsample period before the SEC allowed firms to disclose financials via their websites instead of via EDGAR (except in the most demanding firm and time effects specification). This robustness suggests that the negative relation between our website-size measure and firms' liquidity does not merely reflect firms' disclosure of mandatory filings via their websites (Levy and Taylor [2022]).

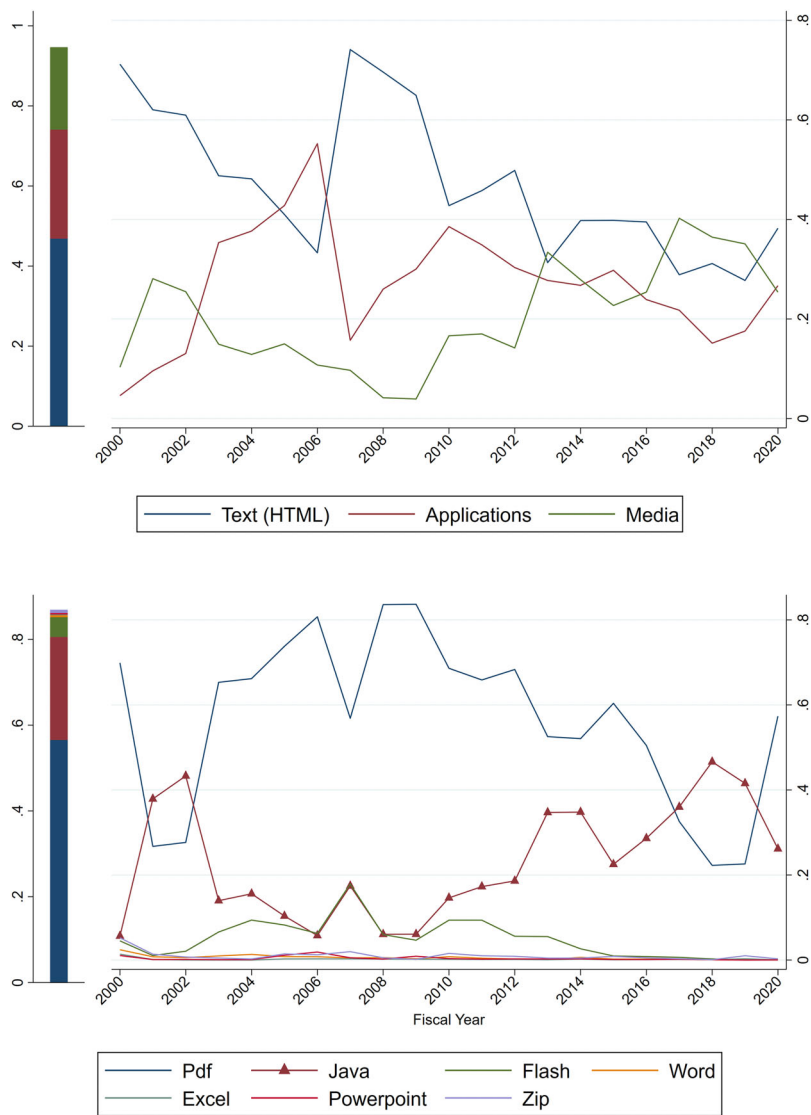


FIG. 3.—Website components. The figures plot the components of websites using our sample of U.S. public firms. The top panel plots the three main components of company websites, while the bottom panel plots a breakdown of the main application formats used on the websites. Each panel presents the average share for the entire period (bars to the left) and their time-series variation (lines to the right) for the period from 2000 to 2020. We compute firm-level fractions of each component at year-end as a proportion of the total website size (top panel) or as a proportion of total application size (bottom panel). Firm-level fractions are averaged across firms on a year-by-year basis.

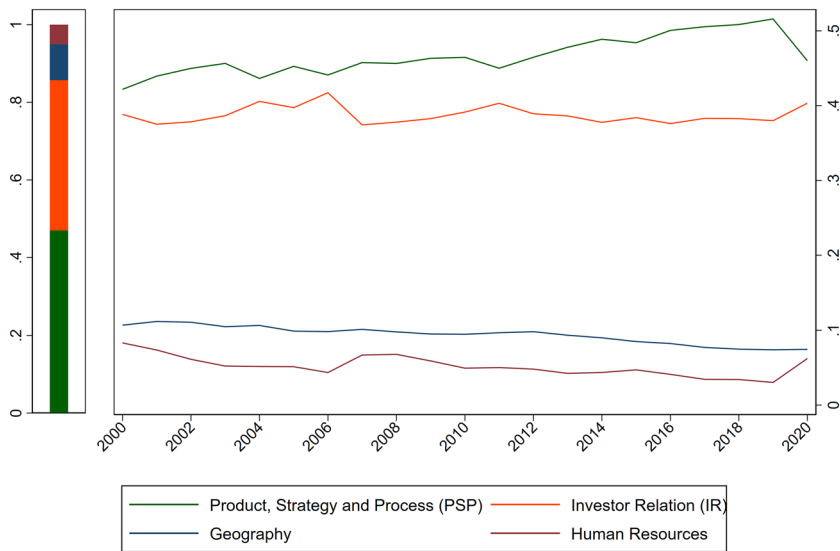


FIG. 4. —Website contents. The figure plots the four main content categories of company websites. We classify web content based on words in the URL string into one of four content categories: *Product, Strategy and Process (PSP)*; *Geography*; *Investor Relations (IR)*; and *Human Resources*. The words used for each category are listed in appendix C. We present the average share for the entire period (bars to the left) and their time-series variation (lines to the right) for the period from 2000 to 2020. We compute firm-level fractions of each content category at year-end as a proportion of the total website size that can be classified in one of the four topics. Firm-level fractions are averaged across firms on a year-by-year basis.

with content on firms’ production, products, innovation, customers, and suppliers. The “Investor Relations” (IR) content category comprises URLs with information on firms’ financials and corporate governance. The “Geography” content category comprises URLs with information on firms’ locations (domestic and foreign), which is useful for understanding the scope of firms’ activities. Finally, the “Human Resources” content category encompasses URLs with content related to employment opportunities and workforce practices.¹⁶ We compute firm-level fractions of each content category at year-end as a proportion of the total website size that can be classified into one of the four topics.

¹⁶ In addition, we categorize words in the following categories: (1) “Communication,” which encompasses all items related to the press, social media, events, public relations, and announcements; and (2) “Corporate Social Responsibility,” which encompasses words related to environment, diversity, and social impact. As those two categories represent a small share of firms’ websites, we exclude them from our analysis. The full list of words used for our four content categories is shown in appendix A. An Excel file containing the full list of words for the six categories, along with the code to build the content measure is available in the replication package.

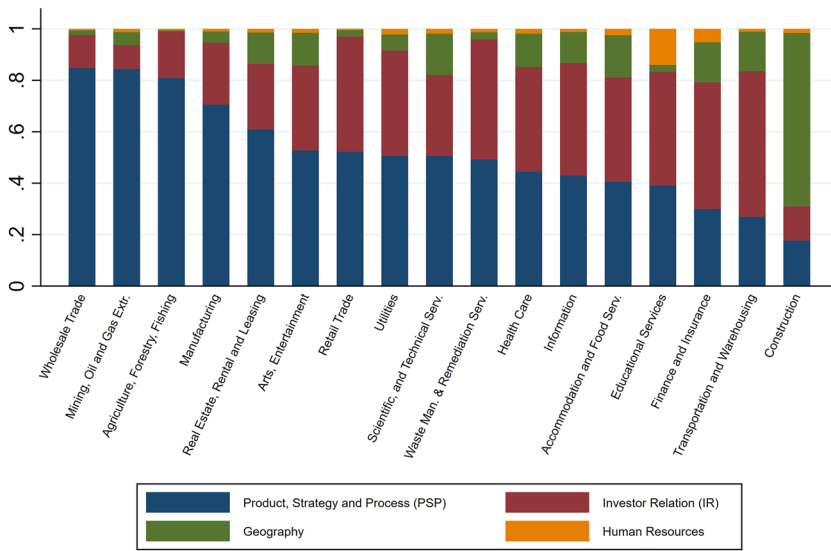


FIG. 5.—Website contents by industry. The figure presents the relative share of content categories of firms’ websites based on their URL strings for our sample of public firms. We group firms into 17 industries based on their NAICS code.

The figure documents that public firms predominantly inform the public of their product, strategy, and processes, consistent with websites targeting stakeholder such as customers and suppliers. It also shows, however, that investor-related contents make up a sizable share of firms’ websites. This finding aligns with our earlier finding that firms’ website contents are associated with extant financial disclosure measures and investors’ information asymmetry. The prominence of investor-related contents on public firms’ websites also aligns with prior literature suggesting that these firms communicate their financial information frequently and in a timely manner with investors on their websites (e.g., Levy and Taylor [2022]). Besides product and investor-related contents, firms also talk about their geographic reach and human resource activities, albeit to a lesser extent. Those additional content categories reinforce the view that websites provide both financial and nonfinancial information relevant to a wide range of stakeholders, including investors but also customers, suppliers, and employees.

The relative frequency of the content categories discussed on firms’ websites does not vary much over time (figure 4) but shows notable differences across industries (figure 5). Product-related issues, for example, are prominently discussed in industries such as agriculture and manufacturing. By contrast, investor-related issues make up much of firms’ website contents in the finance industry. And human-resource-related issues constitute a large fraction of firms’ website content in the education sector, while firms’ geography (e.g., their store locations) takes up a sizable fraction of firms’ website content in the food and accommodation services sector. Broadly,

this cross-industry variation appears to align with intuitive differences in the nature of the industries' main assets, the geographical scale of their operations, and the importance of labor.

In sum, our exploration of website content suggests that company websites contain information on central aspects of firms' businesses, communicated via a range of components and application formats. The disclosed information is likely relevant for various stakeholders, including investors, customers, suppliers, and employees. Consistent with firms' disclosing information specific to their businesses and relevant to their stakeholders, we observe systematic differences in website contents between industries. We explore the systematic factors explaining firms' website size, components, and contents in the next section.

4.2 ASSOCIATED FACTORS

Table 5 explores factors explaining firms' website disclosures. Panel A dissects systematic factors driving variation in firms' website size, our main measure of website-based disclosure. It documents that common industry and time factors explain only a relatively small amount of the website-size variation (about 4% and 12%, respectively). Notably, the limited explanatory power of time suggests that variation in our website-size measure does not merely reflect a time trend toward greater Internet and website use. In contrast to common industry and time factors, fixed firm factors explain a large part of our disclosure measures (about 42%). Regressions of our disclosure measure on firm fixed effects explain roughly four (ten) times more variation in our disclosure measures than regressions using time (industry) fixed effects. We draw similar conclusions for the underlying components and contents of firms' websites. A somewhat different picture emerges only when we examine the relative share of firms' website components (e.g., text vs. images). Here, we observe that time factors explain a sizable fraction of the variation, consistent with the notable time trends in figure 3.

The variance decomposition supports the notion that our measure captures systematic variation in firms' voluntary disclosure decisions. While the website components (e.g., text vs. images) used to communicate firms' information vary over time, the total amount of firms' disclosure on websites appears relatively sticky over time. This feature of our website-based measure could arise, in part, because many aspects of firms' business are comparably persistent (e.g., their product portfolio) and that some degree of stability in the content on and delivery of firms' websites presumably helps stakeholders (e.g., customers) to navigate the website. The persistent differences in website content across firms further aligns with the fact that firms tend to commit to a particular disclosure strategy (Leuz and Verrecchia [2000]). Our variance decomposition, however, also shows that, even after we account for common firm and time factors, a notable fraction of our website-based measure's variation remains unexplained. Accordingly, our measure cannot only be used to study across firm differences but also lends its to studying within-firm changes in disclosure incentives via event

TABLE 5
Determinants of Website Disclosure

Panel A: Variance decomposition			
Website Size	Industry (GICS)	Time	Firm
Industry (GICS)	0.043	-	-
Year-quarter	0.177	0.122	-
Firm	-	0.569	0.416
%(HTML)			
Industry (GICS)	0.018	-	-
Time	0.228	0.209	-
Firm	-	0.387	0.197
%(Media)			
Industry (GICS)	0.007	-	-
Time	0.168	0.163	-
Firm	-	0.289	0.150
%(Applications)			
Industry (GICS)	0.032	-	-
Time	0.164	0.131	-
Firm	-	0.346	0.217
%(PSP)			
Industry (GICS)	0.151	-	-
Time	0.161	0.006	-
Firm	-	0.568	0.550

(Continued)

TABLE 5—(Continued)

TABLE 5—(Continued)

	Panel B: Determinants								
	Website Size			Website Components			Website Content Categories		
	(1)	(2)	(3)	%(HTML)	%(Media)	%(Applications)	%(PSP)	%(IR)	%(Other)
<i>Return Variability</i>	5.995*** (1.464)	0.766 (1.415)	2.360** (1.080)	-0.045 (0.167)	-0.390*** (0.106)	0.377** (0.152)	1.602*** (0.192)	-1.535*** (0.185)	-0.067 (0.136)
<i>Segments</i>	-0.220*** (0.047)	-0.190*** (0.047)	-0.109** (0.049)	-0.004 (0.005)	-0.008** (0.003)	0.009 (0.004)	-0.004 (0.006)	0.014** (0.005)	-0.011** (0.004)
<i>Geographical Complexity</i>	-0.165*** (0.057)	-0.017 (0.058)	0.152*** (0.064)	0.017*** (0.006)	-0.002 (0.004)	-0.017*** (0.005)	0.007 (0.007)	-0.024*** (0.007)	0.017*** (0.005)
<i>Restructuring</i>	0.476*** (0.056)	0.266*** (0.054)	0.112*** (0.038)	-0.018*** (0.006)	-0.006* (0.004)	0.030*** (0.006)	0.044*** (0.006)	-0.027*** (0.006)	-0.017*** (0.004)
<i>Business-to-Business</i>	0.421*** (0.076)			-0.035*** (0.008)	-0.015*** (0.005)	0.060*** (0.006)	0.145*** (0.009)	-0.124*** (0.009)	-0.021*** (0.007)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	No	Yes	No	No	No	No	No	No	No
Firm FE	No	No	Yes	No	No	No	No	No	No
Observations	37,672	37,672	37,672	36,527	36,527	36,527	34,918	34,918	34,918
Adjusted R ²	0.147	0.185	0.520	0.151	0.157	0.128	0.068	0.052	0.022

This table presents determinants of our firms' website disclosure for our sample of U.S. public firms. In panel A, we report R-squares of quarterly regressions of *Website Size*, website component shares (in %), and website content shares (in %) on various time, industry, and year-quarter fixed effects combinations for our sample of public firms. In panel B, we report estimates of OLS regressions of determinants of *Website Size* and various website component and content shares at the annual level. Website components include *%(HTML)* (i.e., the size of text elements (HTML) as a fraction of the total website size), *%(Media)* (i.e., the size of images and videos as a fraction of the total website size), and *%(Applications)* (i.e., the size of applications as a fraction of the total website size). Website contents include *%(PSP)* (i.e., the fraction of the website devoted to the topic "Product, Strategy, Processes"), *%(IR)* (i.e., the fraction of the website devoted to the topic "Investor Relations"), and *%(Other)*, which captures the remaining content of the website. In all regressions, standard errors, reported in parentheses below the coefficient estimates, are clustered at the firm level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in appendix B.

study or difference-in-differences research designs (see, e.g., figure OA1 in the online appendix).

Table 5, panel B, explores specific determinants of firms' website disclosure. After controlling for common time trends, we observe in column 1 that firms with higher revenues, employment, volatility, analyst following, and institutional ownership have larger websites. These findings are consistent with our website measure capturing disclosure provided in response to information demand that arises due to the firms' volatile operations and emanates from large firms' dispersed stakeholders (e.g., Buzby [1975], Breuer, Hombach, and Müller [2023]). These stakeholders include analysts and institutional owners, who are known to encourage public disclosure (e.g., Lang and Lundholm [1996], Boone and White [2015]). We also observe that firms with higher industry-adjusted performance (return on assets), more debt financing (leverage), and more segments have *smaller* websites. These findings are in line with prior literature documenting that profitable and highly levered firms are concerned about the proprietary costs of disclosure (e.g., Berger and Hann [2007], Bernard [2016]) and that multi-segment firms can reduce those costs by aggregating information (e.g., Bens, Berger, and Monahan [2011], Chen, Miao, and Shevlin [2015]). Lastly, we observe that firms undergoing a restructuring and those operating in business-to-business industries have larger websites. The former is consistent with restructuring firms being compelled to provide more detail (Chen, Miao, and Shevlin [2015]). The latter is consistent with firms using websites to communicate with customers and suppliers, especially when channels such as TV advertising and press releases are less efficient (e.g., due to a narrow target audience).

Columns 2 and 3 of table 5, panel B, repeat the determinants analysis with more restrictive fixed effects. Comparing the results to those in column 1, we observe similar associations when focusing only on within-industry variation (column 2) or even just within-firm variation (column 3).¹⁷ While some associations are no longer statistically significant, most retain the same sign as in column 1. The main difference in terms of signs is that the associations of leverage and geographical complexity with firms' websites turn positive when we focus on within-firm variation (column 3). These sign switches may indicate that, while more levered and geographically complex firms generally communicate less via their websites (e.g., because they rely on private information exchange with banks or local consumer contact), a given firm's increased leverage and complexity require more description (e.g., due to increased distress or more stores to list on the website).

In untabulated tests, we also observe similar associations between the determinants and the size of individual components (e.g., text, applications,

¹⁷ We drop the business-to-business industry indicator from the specification with industry or firm fixed effects, as those effects (almost perfectly) subsume the indicator.

and media) and contents (e.g., product, investor, and other) of firms' websites. Given this similarity, we view our measure of firms' total website size as a useful summary measure for the information provided on firms' websites. It is nevertheless interesting to explore how the importance of individual components and contents, relative to the total size of the website, relates to firm characteristics and events. We examine this question next.

Columns 4–9 of table 5 explore the determinants of the *relative* importance of firms' website components and contents. In columns 4 to 6, we observe that firms shift from text (HTML) and/or media (images, videos) toward applications (e.g., PDF files) when they have more revenue, are less levered, are less geographically complex, are restructuring, and are operating in business-to-business industries. In columns 7–9, we further observe that firms focus more on product-related information when they have volatile businesses, are restructuring, and operate in business-to-business industries. We also observe that firms with more employees and geographical complexity appear to talk more about other information, which makes sense given that the “other” category includes information on human resource practices and the geographic location of firms' stores.

On the whole, our results on the determinants of firms' website size and the relative importance of individual components and contents highlight that our website-based disclosure measure varies predictably with firms' fundamentals and stakeholders' information demand. The results document that our measure captures both firms' fundamentals and their decision to disclose those fundamentals. This characteristic is a common feature of disclosure measures, especially those measuring firms' disclosure quantity.

5. Applications

In this section, we showcase the usefulness of our website-based measure for research on private firms and disclosure of nonfinancial information, two increasingly important settings.

5.1 PRIVATE FIRMS

Our website-based disclosure measure promises to be useful for studying private firms. For these firms, extant disclosure measures are often missing, partly because these firms tend to have lower incentives to disclose information publicly (e.g., Buzby [1975], Breuer, Hombach, and Müller [2023]). The lack of disclosure measures, however, also reflects the fact that for private firms, there is usually no centralized database or disclosure platform such as EDGAR that allows researchers to easily measure firms' disclosure activity (e.g., Chuk, Matsumoto, and Miller [2013]).

Our website-based measure can be constructed for private firms. Many company databases (e.g., Orbis, Dun & Bradstreet) often lack private firms' financial information but contain their website URLs. Absent centralized disclosure platforms and dedicated information intermediaries such as analysts, private firms can be expected to rely on their company websites to

communicate with the public. Following this reasoning, we extend our preceding analysis to the realm of private firms' website disclosure, including its determinants and associated capital-market outcomes. One relevant capital-market outcome is private firms' propensity to engage in a private equity transaction. The association between firms' website disclosure and private equity transactions closely resembles the association between public firms' website disclosure and their equity shares' liquidity (i.e., bid-ask spreads). While this association is often studied in the public-firm setting, it has received little attention in the private-firm setting due to the absence of large-scale measures of disclosure. In addition, our website-based measure is plausibly one of the main information sources for firm outsiders (e.g., private equity investors) in the private-firm setting, especially in the U.S.

We present descriptive statistics for our U.S. private-firm sample in panel A of table 6. Our sample is based on a cross-section of several million firms with nonmissing information on revenue, employees, and website URLs in Bureau van Dijk's Orbis (Historical and WRDS) databases (panel A). We limit our sample to a cross-section of private firms as of 2018/2019 because key variables (e.g., revenue and employees) are not available in Orbis before. The scarcity of financial information for U.S. private firms reflects the absence of financial disclosure mandates for private firms in the U.S. It also highlights the promise of websites as a rare source of public information on those firms. After we drop subsidiaries, foreign firms, nonprofit organizations, sole proprietorships, and partnerships, our sample comprises about 3.7 million private firms. For 76% of those firms, we are able to obtain records of their websites from the Wayback Machine. We supplement these data with information on equity transactions from Bureau van Dijk's Zephyr database. We focus on completed deals, including acquisitions of minority and 100% stakes of private firms' equity. To home in on equity transactions by firm outsiders, who are not privy to insider information, we exclude equity transactions resulting from management buy-outs. Our final sample includes over 2.8 million private firms. The firms are predominantly small, with an average (median) of 14 (4) employees (panel A). Recorded equity transactions are rare. Only 0.058% of our sample firms experience an equity transaction in 2019.

We begin our analysis of private firms' website disclosure by exploring the determinants of private firms' website disclosure (panel B of table 6). Our exploration is limited to two firm-size measures, revenue and employment, as financial data on private firms are scarce. Still, as in the public-firm sample, we observe that firms with higher revenue exhibit larger website sizes (column 1), a lower share of text (column 2), and a higher share of applications (column 4) on their websites. Consistent with the results in the public-firm sample, we also observe that firms with more employees tend to exhibit a higher share of content in the "Other" category, which comprises issues related to the human resource content category. We also observe some diverging results. The number of employees, for example, is not significantly associated with firms' website size in the private-firm sam-

TABLE 6
Application to Private Firms

Panel A: Descriptive statistics							
	Obs.	Mean	SD	P25	P50	P75	
<i>Website-based disclosure variable:</i>							
<i>Website Size</i>	2,854,113	10.931	2.891	8.694	10.901	13.124	
<i>Capital market variable:</i>							
<i>Equity Transaction</i> ($\times 100$)	2,854,113	0.058	2.408	0.000	0.000	0.000	
<i>Determinant variables:</i>							
Employment (Count)	2,846,705	14.275	3037.155	3.000	4.000	8.000	
<i>Employment</i>	2,846,705	1.561	1.037	1.099	1.386	2.079	
<i>Revenue</i>	2,836,100	13.254	1.130	12.758	12.986	13.700	
<i>Website components variables:</i>							
% (<i>HTML</i>)	2,775,306	0.610	0.380	0.211	0.717	1.000	
% (<i>Media</i>)	2,775,306	0.267	0.334	0.000	0.062	0.532	
% (<i>Applications</i>)	2,775,306	0.123	0.218	0.000	0.013	0.139	
<i>Website content variables:</i>							
% (<i>PSP</i>)	930,488	0.569	0.346	0.285	0.600	0.912	
% (<i>IR</i>)	930,488	0.278	0.304	0.000	0.182	0.450	
% (<i>Other</i>)	930,488	0.153	0.240	0.000	0.027	0.216	
Panel B: Determinants							
	Website Components				Website Content Categories		
	<i>Website Size</i> (1)	% (<i>HTML</i>) (2)	% (<i>Media</i>) (3)	% (<i>Applications</i>) (4)	% (<i>PSP</i>) (5)	% (<i>IR</i>) (6)	% (<i>Other</i>) (7)
<i>Revenue</i>	0.148*** (0.009)	−0.012*** (0.001)	0.006*** (0.001)	0.006*** (0.001)	0.007*** (0.001)	−0.006*** (0.001)	−0.001 (0.001)
<i>Employment</i>	0.005 (0.013)	−0.001 (0.001)	−0.002** (0.001)	0.003*** (0.001)	−0.007*** (0.001)	0.001 (0.001)	0.006*** (0.001)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,830,160	2,775,306	2,775,306	2,775,306	930,488	930,488	930,488
Adjusted R^2	0.015	0.014	0.008	0.017	0.111	0.068	0.060
Panel C: Equity transactions							
	<i>Private Equity Transaction</i>						
	(1)	(2)	(3)	(4)	(5)		
<i>Website Size</i>	0.018*** (0.003)	0.017** (0.003)	0.015*** (0.003)	0.015*** (0.003)	0.015*** (0.003)	0.015*** (0.003)	0.015*** (0.003)
<i>Revenue</i>			0.106*** (0.012)				0.070*** (0.009)
<i>Employment</i>				0.105*** (0.012)			0.049*** (0.008)
Industry FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,854,111	2,854,111	2,836,097	2,847,703	2,847,703		2,830,160
Adjusted R^2	0.001	0.002	0.004	0.004	0.004		0.004

(Continued)

TABLE 6—(Continued)

This table presents estimates for a sample of over 2.8 million unique U.S. private firms. In panel A, we present descriptive statistics for this sample. In panel B, we present determinant analyses of *Website Size* and various website components and contents. Website components include %(*HTML*) (i.e., the size of text elements (HTML) as a fraction of the total website size), %(*Media*) (i.e., the size of images and videos as a fraction of the total website size), and %(*Applications*) (i.e., the size of applications as a fraction of the total website size). Website contents include %(*PSP*) (i.e., the fraction of the website devoted to the topic “Product, Strategy, Processes”), %(*IR*) (i.e., the fraction of the website devoted to the topic “Investor Relations”), and %(*Other*), which captures the remaining content of the website. In panel C, we present estimates of OLS regressions of the probability of an equity transaction in year 2019 (in percent; i.e., multiplied by 100) on our website-based measure of disclosure in year 2018. Industry fixed effects are based on the four-digit NAICS industry classification. Standard errors, reported below the coefficient estimates, are clustered at the four-digit industry level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in appendix B.

ple. This may be because the revenue and employee numbers of private firms are often estimated by data vendors such as Bureau van Dijk, which reduces their independent variation. We also observe that private firms with higher revenue exhibit a higher share of product, strategy, and processes-related information and a lower share of investor-related information. In the public-firm sample, by contrast, these shares are not significantly associated with firms’ revenue. These differences may hint at some heterogeneous responses across private and public firms. However, they may also simply reflect the fact that we cannot control for the same extensive set of firm characteristics in the private-firm analysis.

We further explore the similarities and differences between private and public firms’ websites in the online appendix. In panel A of table OA1 in the online appendix, we observe that private firms tend to have simpler websites, with more text and less media or applications. These differences make intuitive sense given the stark size differences between the average private firm and the average public firm. In terms of contents, private firms tend to disclose relatively more about their product, strategy, and process and less about investor relations. These differences also make intuitive sense given that private firms, by their very nature, have a more concentrated investor base that often obtains information through private rather than public channels (e.g., Kang [2025]). Despite these differences in the extent to which certain content categories are discussed on private vs. public firms’ websites, we also observe that the website components and contents of private firms vary across industries in ways similar to those of public firms (figures OA2 and OA3 in the online appendix).¹⁸

We next examine whether private firms’ website disclosure is associated with future private-equity transactions. Panel C of table 6 presents results from regressions of our website-based measure (as of 2018) on an indicator

¹⁸In table OA1, panel B, of the online appendix, we explicitly test whether the cross-industry variation in content shares varies significantly across private and public firms. We rank contents based on their prominence in each industry for public and private firms separately. Chi-squared tests fail to reject the null that the relative prominence of contents across industries is the same for private and public firms.

for an equity transaction (in 2019). For ease of presentation and interpretation, we multiply the transaction indicator by 100, expressing it in percentage points. This adjustment is useful given the rarity of (recorded) equity transactions. We observe that our measure exhibits a positive and statistically significant association with the occurrence of an equity transaction (column 1). This association is robust to controlling for industry effects (column 2), number of employees (column 3), and/or revenue (columns 4 and 5). In the online appendix, we also document that the association is robust to using a size-matched subset of U.S. private firms (table OA2 in the online appendix). It also holds when we use a panel of private firms in the United Kingdom (U.K.). Because these firms are subject to the U.K.'s financial disclosure mandate, we can control more extensively for potentially confounding differences across firms by including financial controls and firm fixed effects (table OA3 in the online appendix). As in the public-firm context, we again find that the underlying components and contents are associated with private equity transactions.

The robust association between firms' website size and private equity transactions is consistent with firms' public disclosure via websites alleviating investors' information issues (including information asymmetry and attention issues; Merton [1987], Verrecchia [2001], Blankespoor, deHaan, and Marinovic [2020]). In terms of economic magnitude, the coefficient estimates imply only small changes in firms' likelihood of experiencing an equity transaction. A 10% increase in website size is roughly associated with a 0.0015 percentage point increase in the likelihood of an equity transaction (column 5). This small magnitude reflects the low base rate of recorded equity transactions. Relative to the mean of 0.058%, it translates into a notable 2.58% increase in equity transactions in response to a 10% increase in firms' website size.¹⁹

In sum, this application of our measure to the private-firm setting provides novel evidence that private firms' website disclosure can facilitate private-equity transactions. Our evidence is consistent with private firms using company websites to attract funding, in line with the conjecture in Hope and Vyas [2017]. More broadly, our private-firm application illustrates that in many respects, private firms' website disclosures behave similarly to public firms' disclosure; for example, both types of disclosure seem to help firms attract investors. Private and public firm disclosures are not the same, though, as private firms differ in important ways from public firms (e.g., with respect to their investor orientation), and these differences are reflected in the firms' disclosure strategies and contents (e.g., the emphasis on product over investor/financial information; Kang [2025]). This insight suggests that it is important to examine the disclosure of private firms specifically, especially given the sheer number of private firms in the

¹⁹ The elasticity of equity transactions with respect to firms' website size implied by our estimate in column 5 is $2.58\%/10\% = 0.258$.

economy and the growing size of private capital markets (e.g., Bauguess, Gullapalli, and Ivanov [2018], Bernstein [2022], Ewens and Farre-Mensa [2022]).

5.2 NONFINANCIAL DISCLOSURE

Our website-based disclosure measure also promises to be useful for studying nonfinancial disclosure. Firm stakeholders and regulators increasingly demand information from firms about their nonfinancial performance. For example, this information can help environmentally conscious customers identify sustainable producers (e.g., Darendeli et al. [2022], Beyer et al. [2024]) and help employees search for fair and equitable employers (e.g., Choi et al. [2023]). Notably, this information is relevant to many stakeholders beyond investors in public capital markets; this is why such information is often demanded regardless of firms' listing status. As a result, company websites, which allow firms to communicate financial and nonfinancial information to various stakeholders, are poised to become ever more important in firms' disclosure choices and strategies.

We illustrate the usefulness of our measure for studying the disclosure of nonfinancial information in the context of a gender-equality disclosure mandate in France. Since 2019, both public firms and larger private firms must report a Gender Equality Index. The index combines various proxies for gender equality within a firm's workforce, including gender gaps in pay and promotions. The Department of Labor publishes the index scores on a dedicated website, allowing interested stakeholders to learn about firms' gender-related workforce practices. To amplify the impact of the index, the Department of Labor explicitly requires that firms also publish their index scores on their company websites. This requirement acknowledges that stakeholders, such as employees, may not be aware of the Department of Labor's dedicated website and are more likely to visit firms' websites (e.g., due to information or attention constraints; Christensen, Floyd, and Maffett [2017], Blankespoor, deHaan, and Marinovic [2020]).

We examine firms' compliance with the Gender Equality Index disclosure mandate. Using the Wayback Machine, we collect historical copies of websites as of 2023 for all publicly listed firms and the largest private firms in France. We use COMPUSTAT Global to obtain public firms' website URLs and Bureau van Dijk's Orbis to obtain private firms' website URLs. We restrict the sample of private firms to firms with nonmissing financial information and more than 250 employees.²⁰ This size cutoff limits our sample to firms classified as "large" by Orbis and several European size-based regulations. Limiting the sample to larger private firms helps us focus on the most relevant firms (employers), ensures that the private firms we examine operate in a comparable regulatory environment, and reduces

²⁰ Originally, the mandate applied to private firms with more than 1,000 employees. Since 2020, the size threshold was lowered to 50 employees.

size differences between the private and public firms in our sample. Our final sample comprises 229 public firms and 1,244 private firms. We search the company website of each firm for mentions of the Gender Equality Index. We create an indicator (*Compliance*) that takes the value of one for firms with a mention of the index and zero otherwise.²¹ For firms with a mention, we also create a continuous variable (*Prominence*) capturing the prominence with which the index score is presented. We proxy for the prominence using (the inverse of) the number of slashes (“/”) in the URL of the page on which the index is mentioned. This proxy considers mentions on the main page(s) of firms’ websites as more prominent than mentions on sub-pages. We complement the website disclosure data with information on firms’ Gender Equality Index score as reported on the Department of Labor’s dedicated website.²²

In panel A of table 7, we document that compliance with the nonfinancial disclosure mandate is imperfect. About 69% of public firms comply with the requirement to disclose the index score on their websites. Compliance is even lower among private firms, with just about 25% of private firms disclosing the score on their websites. In panel B, we explore whether firms’ imperfect compliance is systematically related to firms’ underlying Gender Index score. In column 1, we observe that higher scores are associated with higher levels of compliance with the website-disclosure requirement. This association holds even after we control for firm characteristics (column 2) and industry effects (column 3). In column 4, we further observe that the index scores are positively associated with the prominence with which the index scores are shown on the websites. These associations are again robust to controlling for firm characteristics (column 5) and industry effects (column 6). Collectively, these results suggest that firms engage in strategic disclosure: They are more likely to report their scores when the scores are high. Interestingly, this behavior appears concentrated among private firms. In panel C, we document that public firms’ scores are not significantly associated with their disclosure decisions. By contrast, private firms’ scores are significantly (more) positively related to their disclosure decisions. Accordingly, private firms’ strategic disclosure behavior appears to drive the results documented in panel B.

In sum, this application of our measure provides novel evidence that firms strategically avoid compliance with a recent gender-equality disclo-

²¹ To locate the Index, we search for occurrences of the word “égalité” (French for equality). The word “égalité” should unambiguously refer to the “Indice égalité homme-femme” as it is rarely used in a business context. We check manually that this procedure correctly identifies the webpage where the Gender Equality Index is located.

²² Public firms with multiple establishments are not required to publish a consolidated index across all establishments. To compute a single index for public firms, we match public firms with data from the Department of Labor using string matching; whenever there are several matches, we compute the average score across all establishments. There are 27 public firms with multiple establishments. Our results are robust to excluding these firms from our sample.

TABLE 7
Application to Nonfinancial Information

Panel A: Descriptive statistics						
	Obs.	Mean	SD	P25	P50	P75
Public Firms						
Compliance	229	0.694	0.462	0.000	1.000	1.000
Prominence	159	−3.283	1.351	−4.000	−3.000	−2.000
Gender Equality Index	229	0.867	0.094	0.840	0.890	0.930
Revenue	229	5.705	2.662	3.564	5.655	7.984
Employment	229	7.307	2.492	5.361	6.928	9.400
Private Firms						
Compliance	1,244	0.251	0.434	0.000	0.000	1.000
Prominence	312	−2.548	1.254	−3.000	−2.000	−2.000
Gender Equality Index	1,244	0.872	0.093	0.840	0.885	0.930
Revenue	1,244	3.881	1.085	3.100	3.770	4.436
Employment	1,244	5.276	0.872	4.659	5.130	5.733
Panel B: Gender Equality Index disclosure on websites						
	Compliance			Prominence		
	(1)	(2)	(3)	(4)	(5)	(6)
Gender Equality Index	0.654*** (0.117)	0.590*** (0.112)	0.643*** (0.116)	2.949*** (0.816)	2.681** (0.812)	2.133** (1.028)
Revenue		0.006 (0.018)	0.014 (0.020)		−0.154* (0.082)	−0.192 (0.131)
Employment		0.086*** (0.020)	0.083*** (0.022)		−0.007 (0.081)	0.048 (0.125)
Industry FE	No	No	Yes	No	No	Yes
Observations	1,473	1,473	1,473	471	471	471
Adjusted R ²	0.017	0.101	0.170	0.030	0.082	0.204
Panel C: Gender Equality Index disclosure of public versus private firms						
	Compliance			Prominence		
	(1)	(2)	(3)	(4)	(5)	(6)
Gender Equality Index × Private	0.695** (0.326)	0.766** (0.346)	0.782** (0.377)	3.875*** (1.640)	3.244** (1.605)	3.840** (1.866)
Gender Equality Index	0.103 (0.319)	0.008 (0.347)	0.044 (0.377)	0.445 (1.117)	0.873 (1.122)	0.431 (1.364)
Private	−1.050*** (0.290)	−1.014*** (0.299)	−1.010** (0.320)	−2.767* (1.439)	−2.425* (1.422)	−3.105* (1.703)
Revenue		0.013 (0.016)	0.019 (0.020)		−0.139* (0.079)	−0.171 (0.132)
Employment		0.037** (0.017)	0.036** (0.021)		0.048 (0.081)	0.077 (0.126)
Industry FE	No	No	Yes	No	No	Yes
Observations	1,473	1,473	1,473	471	471	471
Adjusted R ²	0.140	0.159	0.215	0.087	0.102	0.216

(Continued)

TABLE 7—(Continued)

This table presents estimates of French firms’ disclosure of a Gender Equality Index on their websites. Our sample of French firms includes firms publicly listed on Euronext Paris (229 unique firms) and private firms considered “Large” or “Very Large” by Orbis (1,244 unique firms). We search the firms’ websites for the disclosure of the index in 2023. We construct two variables: *Compliance*, which captures firms’ compliance with the mandate to disclose their index on their websites, and *Prominence*, which captures the prominence with which firms present the index on their websites. We proxy for the prominence using (the inverse of) the number of sub-levels (or “/”) between the main company website and the URL where the index is disclosed. We obtain the firms’ Gender Equality Index (for 2022, reported in 2023) from the dedicated website of the French Department of Labor. Panel A presents descriptive statistics. Panel B presents results for the pooled sample of public and private firms. Panel C presents results for public vs. private firms. Financial variables are obtained from Compustat Global and Orbis. They are lagged by a year and winsorized at the 1st and 99th percentile. Industry fixed effects are defined using two-digit NAF codes. Standard errors, reported in parentheses below the coefficient estimates, are clustered at the four-digit industry level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in appendix B.

sure mandate in France. It adds to emerging work on the disclosure and misreporting of nonfinancial information (e.g., Lino et al. [2022], Bailey et al. [2025]). Interestingly, our measure reveals that *non*-compliance is not only more prevalent among private firms but also more strategic. The strategic noncompliance among private firms appears to align with the disclosure behavior predicted by *voluntary* disclosure theories in cases when there is no mandate or imperfect enforcement of a mandate. This finding indicates that enforcement issues may be magnified for mandates that extend to private firms. It also underscores the potential limitations of extrapolating the responses of public firms to the sphere of private firms.

6. Conclusion

Company websites represent an important channel through which firms communicate with the public. Building on this insight, we construct and validate a novel measure of firms’ disclosure using firms’ historic website sizes and contents, archived by the Wayback Machine.

Our website-based measure of disclosure comprises rich cross-sectional and time-series variation in firms’ website size and contents. This variation is systematically related to firms’ underlying activities (e.g., their size and volatility) and provides an opportunity to study firms’ disclosure and disclosed activities. Consistent with our measure capturing firms’ disclosure of relevant information, we observe that our measure is positively associated with extant measures of firms’ disclosure (e.g., management forecasts and voluntary 8-K disclosures) and negatively associated with investors’ information asymmetry, even after we control for various aspects of firms’ underlying activities and characteristics.

Our website-based measure promises to be of use in various research settings thanks to its wide availability and customizability. It holds particular promise for studying the disclosure of private firms for which other measures of disclosure are largely lacking. To showcase this promise, we construct our website-based disclosure measure for millions of U.S. private

firms and document novel evidence consistent with private firms' website disclosure facilitating private equity transactions. Our measure also holds promise for examining disclosure to audiences other than investors (e.g., employees) and for researching the disclosure of nonfinancial information (e.g., ESG activities), two topics of growing importance given increasing concerns about firms' societal and environmental impacts. To showcase this promise, we study firms' compliance with a nonfinancial disclosure mandate in France and document novel evidence that private firms exhibit lower compliance and greater strategic nondisclosure than public firms. Together, our two applications illustrate the vast gap in knowledge about private firms' disclosure that our measure can help fill. They also underscore the importance of filling this gap by demonstrating that private firms do not necessarily behave like public ones.

APPENDIX A: SAMPLE SELECTION

This table in the appendix describes the sample selection process for the three distinct samples used in our main analyses. For the analysis of U.S. public and private firms' websites, we rely on URL features (size, format, and content), which can be collected for a large number of firms from the Wayback Machine API (subsection 2.3.1). For the analysis of French public and private firms' websites, we parse the entire website content, not just the URLs. Due to the time-intensive nature of this task, we first restrict the French sample to firms with control variables available to conduct the analysis.

Sample	Observations
U.S. Public Firms	
Unique firms with available website in the CRSP-COMPUSTAT merged dataset	6,985
Unique firms with a record in Wayback Machine and a successful download	5,617
Unique firms with all available variables to conduct the analysis	5,324
U.S. Private Firms	
Unique firms with available website in Orbis	3,761,092
Unique firms with a record in Wayback Machine and a successful download	2,852,457
Unique firms with all available variables to conduct the analysis	2,830,160

Sample	Observations
French Public Firms	
Unique firms with available website in Compustat identification file	519
Unique firms with all available variables to conduct the analysis	371
Unique firms with a record in Wayback Machine and a successful download	229
French Private Firms	
Unique firms with available website from the “Very large” or “Large” categories in Orbis	3,472
Unique firms with all available variables to conduct the analysis	1,490
Unique firms with a record in Wayback Machine and a successful download	1,244

APPENDIX B: VARIABLE DEFINITIONS

This table in the appendix provides variable definitions and data sources for the variables constructed for our sample of U.S. public firms, U.S. private firms, and French firms.

Panel A: U.S. Public firms		
Variable Name	Variable Definition	Source
<i>Website Size</i>	Natural logarithm of the total size of the website in bytes	Wayback Machine
<i>Management Forecast</i>	Natural logarithm of (1+) the number of management earnings forecasts	IBES
<i>Voluntary 8-K Filings</i>	Natural logarithm of (1+) the word count of voluntary 8-K filings (He and Plumlee [2020])	EDGAR
<i>Disclosure Quality</i>	Natural logarithm of (1+) the number of disaggregated items in financial statements (Chen, Miao, and Shevlin [2015])	COMPUSTAT
<i>Gross 10-K File Size</i>	Natural logarithm of (1+) the number of characters in 10-K filings (Loughran and McDonald [2014])	EDGAR
<i>Bid–Ask Spread</i>	Natural logarithm of the quarterly median of the quoted bid–ask spread at the end of each trading day, calculated as the difference between the bid and the ask price divided by the midpoint price	CRSP
<i>Market Value</i>	Natural logarithm of the market value of equity (stock price multiplied by shares outstanding)	CRSP
<i>Share Turnover</i>	Natural logarithm of the average of the daily turnover, measured as the trading volume divided by the market capitalization at the end of each trading day over a quarter	CRSP

Panel A: U.S. Public firms		
Variable Name	Variable Definition	Source
<i>Return Variability</i>	Natural logarithm of the standard deviation of daily stock returns over a quarter	CRSP
<i>Analyst Following</i>	Natural logarithm of the number of analysts following a firm	IBES
<i>Revenue</i>	Natural logarithm of total revenues	COMPUSTAT
<i>Institutional Ownership</i>	Percentage of institutional shareholders from 13-F filings	EDGAR
<i>ROA</i>	Industry-adjusted return on assets calculated as a firm's net income over total assets less the industry's mean return on assets	COMPUSTAT
<i>Leverage</i>	Short-term debt plus long-term debt over total assets	COMPUSTAT
<i>Segments</i>	Natural logarithm of the number of reported business segments	COMPUSTAT
<i>Geographical Complexity</i>	Natural logarithm of the number of states in which firms have operations in 10-K filings (Garcia and Norli [2012])	EDGAR
<i>Restructuring</i>	Indicator variable taking the value of one if the restructuring item is different from zero	COMPUSTAT
<i>Business-to-Business</i>	Indicator variable taking the value of one for industries with supplies to Personal Consumption	Delgado and Mills [2020]
<i>%(HTML)</i>	Expenditure lower than 35% Size of text elements (HTML) as a fraction of the total website size	Wayback Machine
<i>%(Media)</i>	Size of media (images and videos) as a fraction of the total website size	Wayback Machine
<i>%(Applications)</i>	Size of applications as a fraction of the total website size	Wayback Machine
<i>%(PSP)</i>	The fraction of the website devoted to the topic <i>Product, Strategy, Processes</i> (PSP)	Wayback Machine
<i>%(IR)</i>	The fraction of the website devoted to the topic <i>Investor Relations</i> (IR)	Wayback Machine
<i>%(Other)</i>	The fraction of the website devoted to topics other than <i>PSP</i> or <i>IR</i>	Wayback Machine
Panel B: U.S. private firms		
Variable Name	Variable Definition	Source
<i>Website Size</i>	Natural logarithm of the total size of the website in bytes	Wayback Machine
<i>Private Equity Transaction</i>	Indicator variable taking the value of one if a deal involving a 100% acquisition or a minority stake was completed (multiplied by 100)	Zephyr
<i>Revenue</i>	Natural logarithm of total revenues	Orbis

Panel B: U.S. private firms		
Variable Name	Variable Definition	Source
<i>Employment</i>	Natural logarithm of the total number of employees	Orbis
<i>% (HTML)</i>	Size of text elements (HTML) as a fraction of the total website size	Wayback Machine
<i>% (Media)</i>	Size of media (images and videos) as a fraction of the total website size	Wayback Machine
<i>% (Applications)</i>	Size of applications as a fraction of the total website size	Wayback Machine
<i>% (PSP)</i>	The fraction of the website devoted to the topic <i>Product, Strategy, Processes</i> (PSP)	Wayback Machine
<i>% (IR)</i>	The fraction of the website devoted to the topic <i>Investor Relations</i> (IR)	Wayback Machine
<i>% (Other)</i>	The fraction of the website devoted to topics other than <i>PSP</i> or <i>IR</i>	Wayback Machine
Panel C: French firms		
Variable Name	Variable Definition	Source
<i>Gender Equality Index</i>	Gender Equality Index measured out of 100, rescaled to range from 0 to 1	French Ministry of Labor
<i>Compliance</i>	Indicator taking the value of one if the Gender Equality Index is disclosed publicly on the firm's website, and zero otherwise	Wayback Machine
<i>Prominence</i>	Number of sub-level (or "/") between the main company website multiplied by -1 and the URL where the <i>Gender Equality Index</i> is disclosed	Wayback Machine
<i>Revenue</i>	Natural logarithm of total revenues	Orbis (Private) & Compustat Global (Public)
<i>Employment</i>	Natural logarithm of the total number of employees	Orbis (Private) & Compustat Global (Public)

APPENDIX C: LIST OF KEYWORDS

This table in the appendix lists the keywords used to assign content categories to each URL. To reduce misclassification issues, keywords are not reduced to their root form (i.e., *stemmed*) before they are manually classified. This approach means that several inflections for each word (e.g., plural forms) are contained in the list.

Content Category	Keywords
Product, Strategy and Processes (PSP)	buy, client, clients, consumer, contract, customer, customers, partner, partners, partnership, supply, vendor, partnerships, supplier, suppliers, competition, ads, advertising, brand, brands, launch, launches, marketing, portfolio, price, pricing, product, products, project, purchase, quality, retail, service, services, solution, solutions, campaign, certification, certified, demand, license, licensing, optimization, prices, delivery, building, capacity, data, distribution, energy, equipment, industrial, industry, invest, investment, manufacturing, operations, platform, process, processing, production, protection, container, device, devices, electric, electronic, electronics, engineer, facilities, gas, industries, logistics, machine, oil, operating, operation, pipeline, plant, platforms, platinum, steel, storage, transport, truck, utilities, innovation, research, resource, resources, science, tech, technical, technologies, technology, expert, expertise, ideas, innovations, innovative, intelligence, knowledge, patent, projects, scientific, solar, business, culture, expands, market, markets, mission, shop, strategic, strategy, success, value, values, vision, expand, expansion, opportunity, strategies, drug, stores, store
Investor Relations (IR)	account, asset, assets, audit, benefits, book, cash, sale, sales, accounting, accounts, balance, benefit, billing, books, costs, board, director, directors, employee, employees, executive, governance, head, officer, officers, president, senior, chairman, announce, announces, calendar, earnings, filings, performance, quarter, release, releases, report, reporting, reports, results, schedule, dividend, acquire, acquisition, agreement, analyst, analysts, bank, banking, capital, commercial, companies, company, compliance, corporate, cost, coverage, equity, exchange, finance, financial, financing, growth, insurance, investor, investors, legal, loan, meeting, money, offer, offering, savings, sec, shareholder, statement, stock, target, tax, acquires, announcement, budget, committee, compensation, deal, affiliate, closing, corporation, disclosure, executives, fiscal, fraud, fund, income, investments, law, listing, loans, merger, outlook, ownership, rate, rates, rating, regulatory, result, revenue, shareholders, statements, trading, transaction
Geography	africa, america, american, americas, angeles, arizona, asia, atlanta, austin, australia, boston, brazil, california, canada, canadian, carolina, chicago, china, colorado, columbia, country, county, diego, east, europe, european, florida, france, francisco, french, georgia, german, germany, illinois, india, indiana, italian, italy, japan, jersey, kansas, location, locations, london, mexico, miami, michigan, office, ohio, pacific, paris, philadelphia, santa, seattle, singapore, southern, spain, spanish, state, texas, usa, vegas, virginia, washington, west, western, world, worldwide, york, border, branch, houston, dallas, denver, phoenix, coast, offices, regional career, careers, college, education, employment, experience, job, jobs, opportunities, recruiting, school, students, training, university, campus, student
Human Resources	

REFERENCES

- AERTS, W.; D. CORMIER; I. GORDON; and M. MAGNAN. "Performance Disclosure on the Web: An Exploration of the Impact of Managers' Perceptions of Stakeholder Concerns." *The International Journal of Digital Accounting Research* 6 (2006): 159–94.
- AGHAMOLLA, C., and R. THAKOR. "Do Mandatory Disclosure Requirements for Private Firms Increase the Propensity of Going Public?" *Journal of Accounting Research* 60 (2022): 755–804.
- AKEY, P.; A. ROBERTSON; and M. SIMUTIN. "Noisy Factors? The Retroactive Impact of Methodological Changes on the Fama-French Factors." Rotman School of Management Working Paper, 2023. (SSRN link).
- ANDREICOVICI, I.; N. NIKOLAEV; L. VAN LENT; and R. ZHANG. "Accounting Measurement Intensity." Chicago Booth Research Paper, 2020. (SSRN link).
- ARORA, M., and S. LODHIA. "The BP Gulf of Mexico Oil Spill: Exploring the Link Between Social and Environmental Disclosures and Reputation Risk Management." *Journal of Cleaner Production* 140 (2017): 1287–97.
- BAIK, B.; N. BERFELD; and R. VERDI. "Do Public Financial Statements Influence Venture Capital and Private Equity Financing?" *The Accounting Review* 100 (2025): 1–20.
- BAILEY, M.; S. GLAESER; J. OMARTIAN; and A. RAGHUNANDAN. "Misreporting of Mandatory ESG Disclosures: Evidence from Gender Pay Gap Information." Working paper, 2025. (SSRN link).
- BAUGUESS, S.; R. GULLAPALLI; and V. IVANOV. "Capital Raising in the U.S.: An Analysis of the Market for Unregistered Securities Offerings, 2009–2017." Securities and Exchange Commission White Paper, 2018. (SEC link).
- BEBCHUK, L., and R. TALLARITA. "Will Corporations Deliver Value to All Stakeholders?" *Vanderbilt Law Review* 75 (2022): 1031–91.
- BENS, D.; P. BERGER; and S. MONAHAN. "Discretionary Disclosure in Financial Reporting: An Examination Comparing Internal Firm Data to Externally Reported Segment Data." *The Accounting Review* 86 (2011): 417–49.
- BERGER, P., and R. HANN. "Segment Profitability and the Proprietary and Agency Costs of Disclosure." *The Accounting Review* 82 (2007): 869–906.
- BERNARD, D. "Is the Risk of Product Market Predation A Cost of Disclosure?" *Journal of Accounting and Economics* 62 (2016): 305–25.
- BERNSTEIN, S. "The Effects of Public and Private Equity Markets on Firm Behavior." *Annual Review of Financial Economics* 14 (2022): 295–318.
- BEYER, A.; D. COHEN; T. LYS; and B. WALTHER. "The Financial Reporting Environment: Review of the Recent Literature." *Journal of Accounting and Economics* 50 (2010): 296–343.
- BEYER, B.; R. CHASKEL; S. EULER; J. GASSEN; A.-K. GROSSKOPF; and T. SELHORN. "How Does Carbon Footprint Information Affect Consumer Choice? A Field Experiment." *Journal of Accounting Research* 62 (2024): 101–36.
- BLANKESPOOR, E. "Firm Communication and Investor Response: A Framework and Discussion Integrating Social Media." *Accounting, Organizations and Society* 68–69 (2018): 80–87.
- BLANKESPOOR, E.; E. DEHAAN; and I. MARINOVIC. "Disclosure Processing Costs, Investors' Information Choice, and Equity Market Outcomes: A Review." *Journal of Accounting and Economics* 70 (2020): 101344.
- BLANKESPOOR, E.; G. MILLER; and H. WHITE. "The Role of Dissemination in Market Liquidity: Evidence from Firms' Use of Twitter." *The Accounting Review* 89 (2014): 79–112.
- BOONE, A., and J. WHITE. "The Effect of Institutional Ownership on Firm Transparency and Information Production." *Journal of Financial Economics* 117 (2015): 508–33.
- BOTOSAN, C. "Disclosure Level and the Cost of Equity Capital." *The Accounting Review* 72 (1997): 323–49.
- BREUER, M. "How Does Financial-Reporting Regulation Affect Industry-Wide Resource Allocation?" *Journal of Accounting Research* 59 (2021): 59–110.
- BREUER, M.; K. HOMBACH; and M. MÜLLER. "How Does Financial Reporting Regulation Affect Firms' Banking?" *The Review of Financial Studies* 31 (2017): 1265–97.

- BREUER, M.; K. HOMBACH; and M. A. MÜLLER. "The Economics of Firms' Public Disclosure: Theory and Evidence." Working paper, 2023. (SSRN link).
- BRÜGGEMANN, U.; A. KAUL; C. LEUZ; and I. WERNER. "The Twilight Zone: OTC Regulatory Regimes and Market Quality." *Review of Financial Studies* 31 (2018): 898–942.
- BUCHAK, G.; G. MATVOS; T. PISKORSKI; and A. SERU. "Fintech, Regulatory Arbitrage, and the Rise of Shadow Banks." *Journal of Financial Economics* 130 (2020): 453–83.
- BUSHEE, B.; J. CORE; W. GUAY; and S. HAMM. "The Role of the Business Press as an Information Intermediary." *Journal of Accounting Research* 48 (2010): 1–19.
- L. BUZBY, S.. "Company Size, Listed Versus Unlisted Stocks, and the Extent of Financial Disclosure." *Journal of Accounting Research* 13 (1975): 16–37.
- CHEN, S.; B. MIAO; and T. SHEVLIN. "A New Measure of Disclosure Quality: The Level of Disaggregation of Accounting Data in Annual Reports." *Journal of Accounting Research* 53 (2015): 1017–54.
- CHOI, J.; J. PACELLI; K. RENNEKAMP; and S. TOMAR. "Do Jobseekers Value Diversity Information? Evidence from a Field Experiment and Human Capital Disclosures." *Journal of Accounting Research* 61 (2023): 695–735.
- CHRISTENSEN, H.; E. FLOYD; and M. MAFFETT. "The Real Effects of Mandated Information on Social Responsibility in Financial Reports: Evidence From Mine-Safety Records." *Journal of Accounting and Economics* 64 (2017): 284–304.
- CHRISTENSEN, H.; L. HAIL; and C. LEUZ. "Mandatory IFRS Reporting and Changes in Enforcement." *Journal of Accounting and Economics* 56 (2013): 147–77.
- CHUK, E.; D. MATSUMOTO; and G. MILLER. "Assessing Methods of Identifying Management Forecasts: CIG vs. Researcher Collected." *Journal of Accounting and Economics* 53 (2013): 23–42.
- COLLER, M., and T. YOHN. "Management Forecasts and Information Asymmetry: An Examination of Bid–Ask Spreads." *Journal of Accounting Research* 35 (1997): 181–91.
- CORMIER, D.; M.-J. LEDOUX; and M. MAGNAN. "The Use of Web Sites as a Disclosure Platform for Corporate Performance." *International Journal of Accounting Systems* 10 (2009): 1–24.
- CRENSHAW, C. "Mind the (Data) Gaps." Keynote Address at the 8th Annual Conference on Financial Market Regulation (CFMR), 2021. (SEC link).
- DARENDELI, A.; P. FIECHTER; J.-M. HITZ; and N. LEHMANN. "The Role of Corporate Social Responsibility (CSR) Information in Supply-Chain Contracting: Evidence from the Expansion of CSR Rating Coverage." *Journal of Accounting and Economics* 74 (2022): 101525.
- DELGADO, M., and K. MILLS. "The Supply Chain Economy: A New Industry Categorization for Understanding Innovation in Services." *Research Policy* 49 (2020): 104039.
- DIAMOND, D. "Optimal Release of Information By Firm." *Journal of Finance* 40 (1985): 1071–94.
- DIAMOND, D., and R. VERRECCHIA. "Disclosure, Liquidity, and the Cost of Capital." *Journal of Finance* 46 (1991): 1325–59.
- DRANOVE, D., and G. Z. JIN. "Quality Disclosure and Certification: Theory and Practice." *Journal of Economic Literature* 48 (2010): 935–63.
- DYER, T.; M. LANG; and L. STICE-LAWRENCE. "The Evolution of 10-K Textual Disclosure: Evidence from Latent Dirichlet Allocation." *Journal of Accounting and Economics* 64 (2017): 221–45.
- EGE, M.; J. GLENN; and J. ROBINSON. "Unexpected SEC Resource Constraints and Comment Letter Quality." *Contemporary Accounting Research* 37 (2020): 33–67.
- ETTREDGE, M.; V. RICHARDSON; and S. SCHOLZ. "The Presentation of Financial Information at Corporate Web Sites." *International Journal of Accounting Information Systems* 2 (2001): 149–68.
- EWENS, M., and J. FARRE-MENSA. "Private or Public Equity? The Evolving Entrepreneurial Finance Landscape." *Annual Review of Financial Economics* 14 (2022): 271–93.
- FARMILOE, B. "12 Reasons Why Small Businesses Need a Website." 2023. Available at <https://www.score.org/resource/blog-post/12-reasons-why-small-businesses-need-a-website>.
- GAO, P.; C. LEE; and D. MURPHY. "Financing Dies in Darkness? The Impact of Newspaper Closures on Public Finance." *Journal of Financial Economics* 135 (2020): 445–65.
- GARCIA, D., and O. NORLI. "Geographic Dispersion and Stock Returns." *Journal of Financial Economics* 106 (2012): 547–65.

- GÖZLÜĞÖL, A. A., and W.-G. RINGE. "Private Companies: The Missing Link on the Path to Net Zero." *Journal of Corporate Law Studies* 22 (2022): 887–929.
- HASSAN, T.; S. HOLLANDER; L. VAN LENT; and A. TAHOUN. "Firm-Level Political Risk: Measurement and Effects." *Quarterly Journal of Economics* 134 (2019): 2135–202.
- HE, J., and M. PLUMLEE. "Measuring Disclosure Using 8-K Filings." *Review of Accounting Studies* 25 (2020): 903–62.
- HEFLIN, F., and D. WALLACE. "The BP Oil Spill: Shareholder Wealth Effects and Environmental Disclosures." *Journal of Business, Finance & Accounting* 44 (2017): 337–74.
- HOBERG, G.; C. KNOBLOCK; G. PHILLIPS; J. PUJARA; L. RASCHID; and J. QIU. "Using Representation Learning and Web Text to Identify Competitor Networks." Working paper, 2024. (link).
- HOPE, O.-K., and D. VYAS. "Private Company Finance and Financial Reporting." *Accounting and Business Research* 47 (2017): 506–37.
- HUTTON, A.; G. MILLER; and D. SKINNER. "The Role of Supplementary Statements with Management Earnings Forecasts." *Journal of Accounting Research* 41 (2003): 867–90.
- IR-SOCIETY. "Best Practice Guidelines Corporate Websites Updated March 2012." 2012. Available at https://irsociety.org.uk/files/Best_Practice_Guidelines_Corporate_websites_2012_FINAL.pdf
- KANG, Y. "Financial Reporting Around Private Firms' Securities Offerings." *Journal of Accounting Research* 63 (2025): 857–901.
- KIM, J., and M. OLBERT. "How does Private Firm Disclosure Affect Demand for Public Firm Equity? Evidence from the Global Equity Market." *Journal of Accounting and Economics* 74 (2022): 101545.
- KING, R.; G. POWNALL; and G. WAYMIRE. "Expectations Adjustments Via Timely Management Forecasts: Review, Synthesis, and Suggestions for Future Research." *Journal of Accounting Literature* 9 (1990): 113–44.
- KORETO, R. "When the Bottom Line is Online." *Journal of Accountancy* 183 (1997): 63.
- LANG, M., and R. LUNDHOLM. "Cross-Sectional Determinants of Analyst Ratings of Corporate Disclosures." *Journal of Accounting Research* 31 (1993): 246–71.
- LANG, M., and R. LUNDHOLM. "Corporate Disclosure Policy and Analyst Behavior." *The Accounting Review* (1996): 467–92.
- LEONELLI, S.; M. MUHN; T. RAUTER; and G. SRAN. "How Do Consumers Use Firm Disclosure? Evidence from a Randomized Field Experiment with Everyday Product Purchases." Chicago Booth Research paper, 2025. (SSRN link).
- LESMOND, D.; J. OGDEN; and C. TRZCINKA. "A New Estimate of Transaction Costs." *Review of Financial Studies* 12 (1999): 1113–41.
- LEUZ, C., and R. VERRECCHIA. "The Economic Consequences of Increased Disclosure." *Journal of Accounting Research* 38 (2000): 91–124.
- LEVY, B., and D. TAYLOR. "The Information Content of Corporate Websites." Working paper, 2022. (SSRN link).
- LIM, I.; J. HAGENDORFF; and S. ARMITAGE. "Does Distance Impede Regulatory Monitoring? Evidence from the Banking Industry." Working paper, 2016. (SSRN link).
- LINO, M.; P. DOOLAN; P. DIVGI; and R. MEHROTRA. "Closing the Public–Private Environmental Transparency Gap." Note from Bain & Company, 2022. (link).
- LOUGHRAN, T., and B. McDONALD. "Measuring Readability in Financial Disclosures." *Journal of Finance* 69 (2014): 1643–71.
- MARQUARDT, C., and C. WIEDMAN. "Voluntary Disclosure, Information Asymmetry, and Insider Selling through Secondary Equity Offerings." *Contemporary Accounting Research* 15 (1998): 505–37.
- MAS, A. "Does Transparency Lead to Pay Compression?" *Journal of Political Economy* 125 (2017): 1683–721.
- MATHERLY, M., and H. BURTON. "An Analysis of Corporate Website Disclosures." *Management Accounting Quarterly* 6 (2005): 26–33.
- MERTON, R. "A Simple Model of Capital Market Equilibrium with Incomplete Information." *Journal of Finance* 42 (1987): 483–510.

- MINNIS, M. "Does Greater Private Firm Disclosure Affect Public Equity Markets? A Discussion of Kim and Olbert (2022)." *Journal of Accounting and Economics* 74 (2022): 101543.
- MINNIS, M., and P. LISOWSKY. "The Silent Majority: Private U.S. Firms and the Market for Financial Reporting." *Journal of Accounting Research* 58 (2020): 547–88.
- ORTIZ, M.; C. PETER; F. URZUA; and P. VOLPIN. "Mandatory Financial Disclosure and M&A Activity." *Review of Financial Studies* 36 (2023): 4788–823.
- POLLACH, I. "The Readership of Corporate Websites." *Journal of Business Communication* 48 (2011): 27–53.
- POWNALL, G.; M. VULCHEVA; and X. WANG. "The Creation and Segmentation of the Euronext Stock Exchange: A Solution to the Inadequacy of National Securities Regulators?" *Accounting Horizons* 29 (2015): 853–85.
- ROGERS, J.; D. SKINNER; and S. ZECHMAN. "The Role of the Media in Disseminating Insider-Trading News." *Review of Accounting Studies* 21 (2016): 711–39.
- ROGERS, J.; D. SKINNER; and S. ZECHMAN. "Run EDGAR Run: SEC Dissemination in a High-Frequency World." *Journal of Accounting Research* 55 (2017): 459–505.
- SAUTNER, Z.; L. VAN LENT; G. VILKOV; and R. ZHANG. "Firm-Level Climate Change Exposure." *Journal of Finance* 78 (2023): 1449–98.
- SAXTON, G.; D. NEELY; and C. GUO. "Web Disclosure and the Market for Charitable Contributions." *Journal of Accounting and Public Policy* 33 (2014): 127–44.
- SHE, G. "The Real Effects of Mandatory Nonfinancial Disclosure: Evidence from Supply Chain Transparency." *The Accounting Review* 97 (2022): 399–425.
- VERRECCHIA, R. "Essays on Disclosure." *Journal of Accounting and Economics* 32 (2001): 97–180.
- ZHU, C. "Investor Demand for Contextual Information: Evidence from Wikipedia." Working paper, 2019. (UCLA link).